

INTRODUCTION TO GAMES

Games Research



TODAY'S SESSION

- What is games research?
- Game studies: games as culture and meaning
- Games & HCI: interaction and player experience
- Games user research and UX in industry
- Methods, ethics, and where to learn more



WHAT DO WE MEAN BY "GAMES RESEARCH"?

- Systematic study of games, players, and play contexts
- Combines methods from design, psychology, sociology, and media studies
- **Goal: understand and improve games and play**

WHY STUDY GAMES AT ALL?

- Games are a major cultural and economic medium
- Millions of people play every day
- Games shape identities, communities, and skills



GAMES AS CULTURE, ART, AND TECHNOLOGY

- Games as artworks and stories
- Games as interactive systems and rules
- Games as social spaces and communities



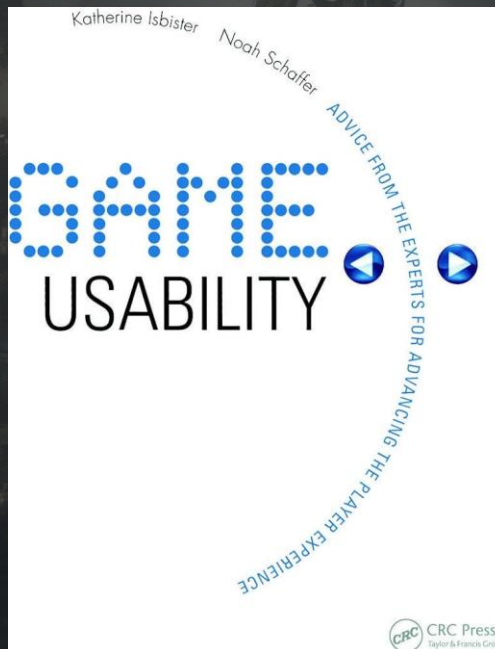
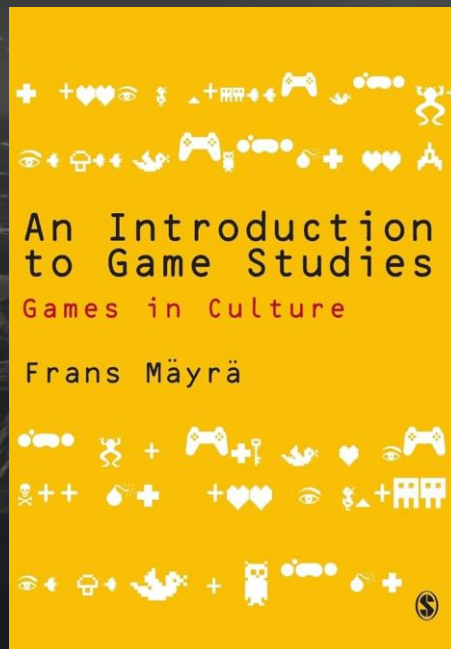
WHERE DOES GAMES RESEARCH HAPPEN?

- Universities and research institutes
- Game studios and UX teams
- Independent scholars and critical writers



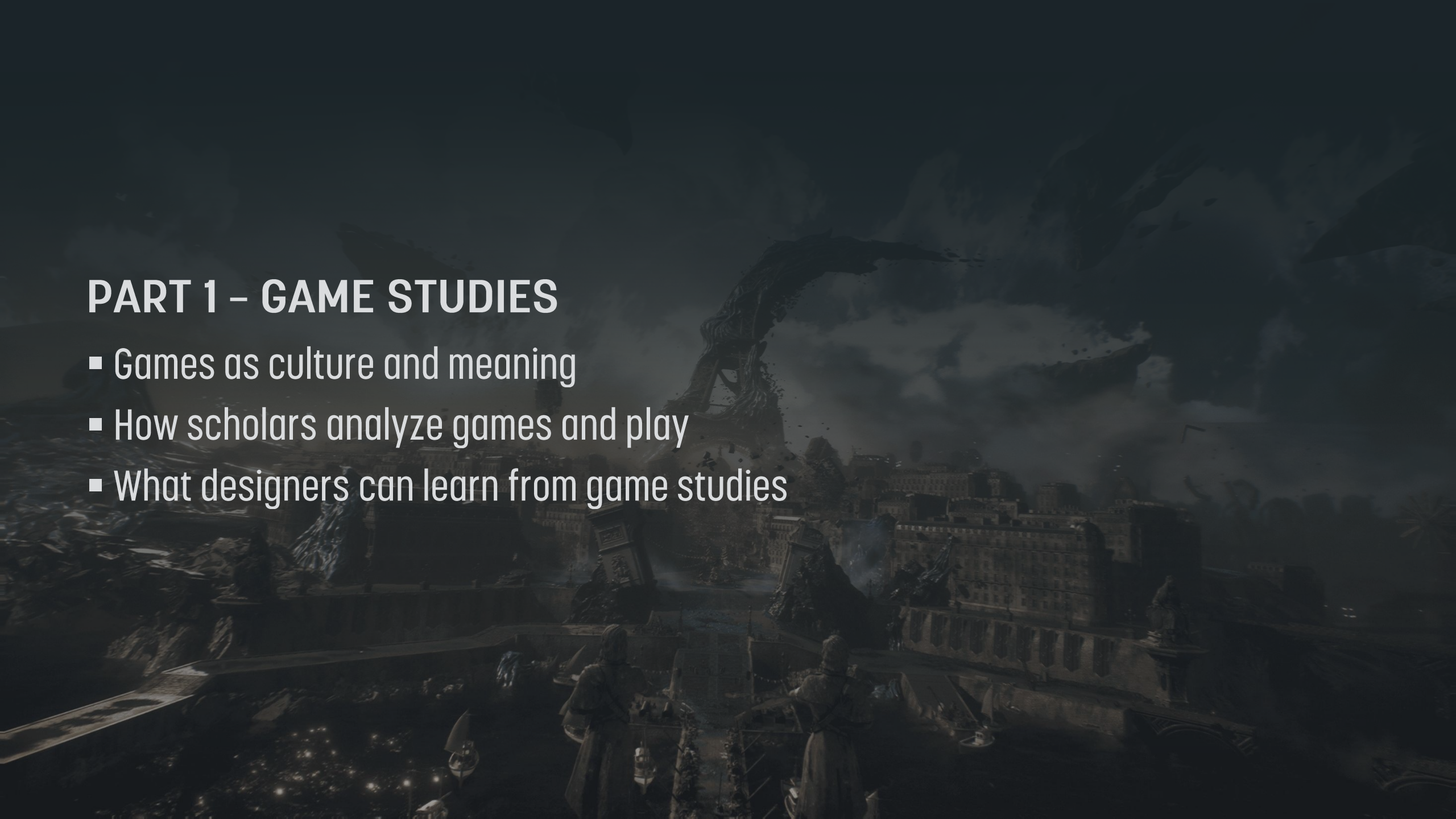
DIFFERENT FLAVORS OF GAMES RESEARCH

- Game studies: games as cultural and symbolic objects
- Games & HCI: interaction, usability, and experience
- Games user research: testing and improving specific games



PART 1 – GAME STUDIES

- Games as culture and meaning
- How scholars analyze games and play
- What designers can learn from game studies



WHAT IS GAME STUDIES?

- Academic field focusing on games, play, and players
- Interdisciplinary: media studies, sociology, anthropology, etc.
- Looks at both digital and non-digital games

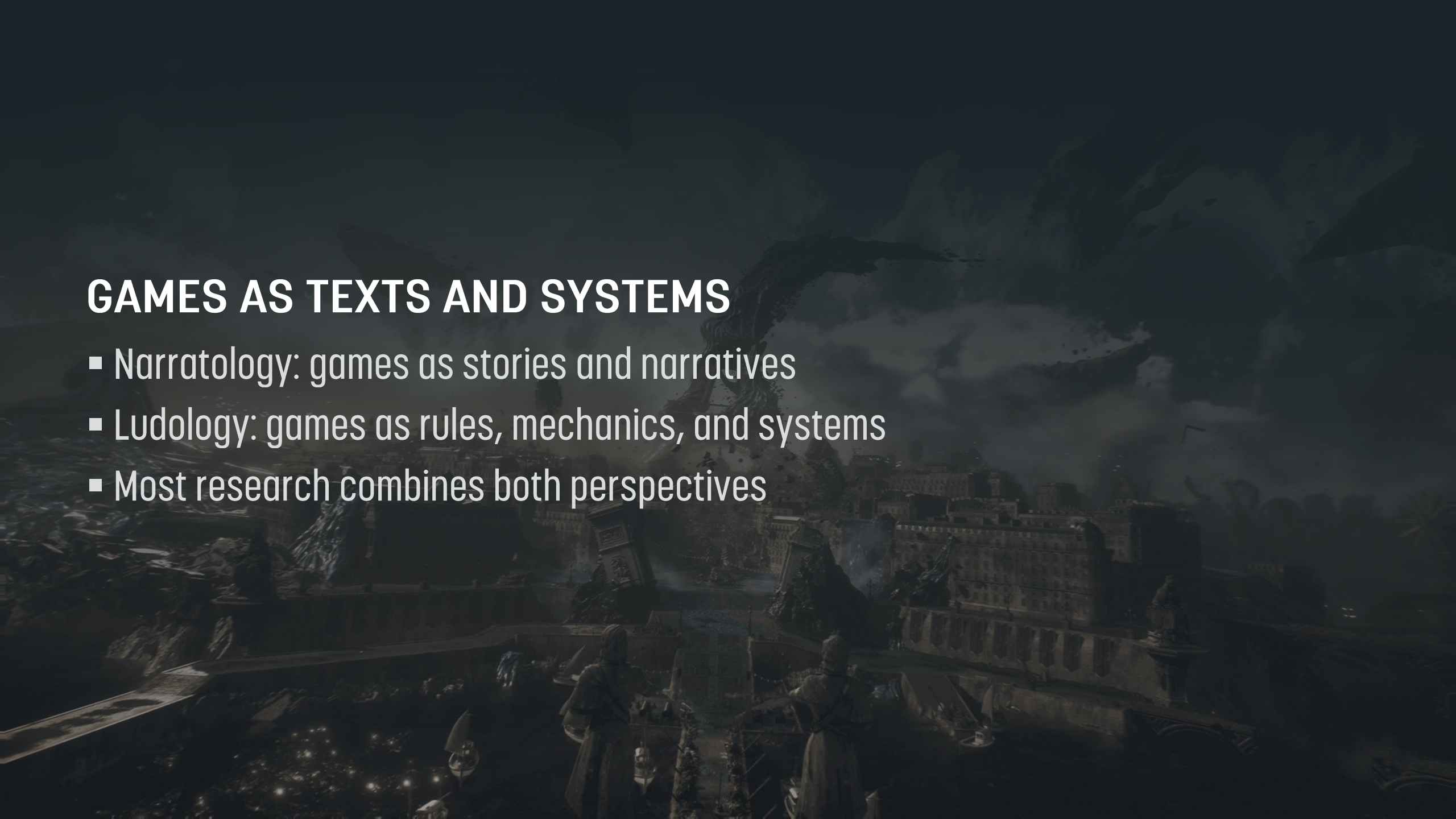


GAME STUDIES?

- **Game studies:** ludology (from ludus, Latin for "game")
- **Scope:** discipline focused on understanding games as multifaceted cultural, social, and technological phenomena
- **Methodologies:** humanities, social sciences, and computer science
- **Research:** how games function as interactive systems, shape human behavior, and reflect broader societal dynamics

GAMES AS TEXTS AND SYSTEMS

- Narratology: games as stories and narratives
- Ludology: games as rules, mechanics, and systems
- Most research combines both perspectives



INTERDISCIPLINARY NATURE

- **Humanities:** Close readings of game texts, historical analysis, and cultural criticism
- **Social Sciences:** Empirical studies on player demographics, psychological impacts, and community dynamics
- **Design and Engineering:** Research on procedural generation, AI, and game mechanics
- **Methodological pluralism:** diverse research questions

CONCEPTUAL FOUNDATIONS

- Early scholarship focused on video games
- Now: encompasses board games, sports, and hybrid digital-physical experiences
- Reflects a recognition of games as universal human practices with historical and cross-cultural significance



KEY TOPICS OF INQUIRY

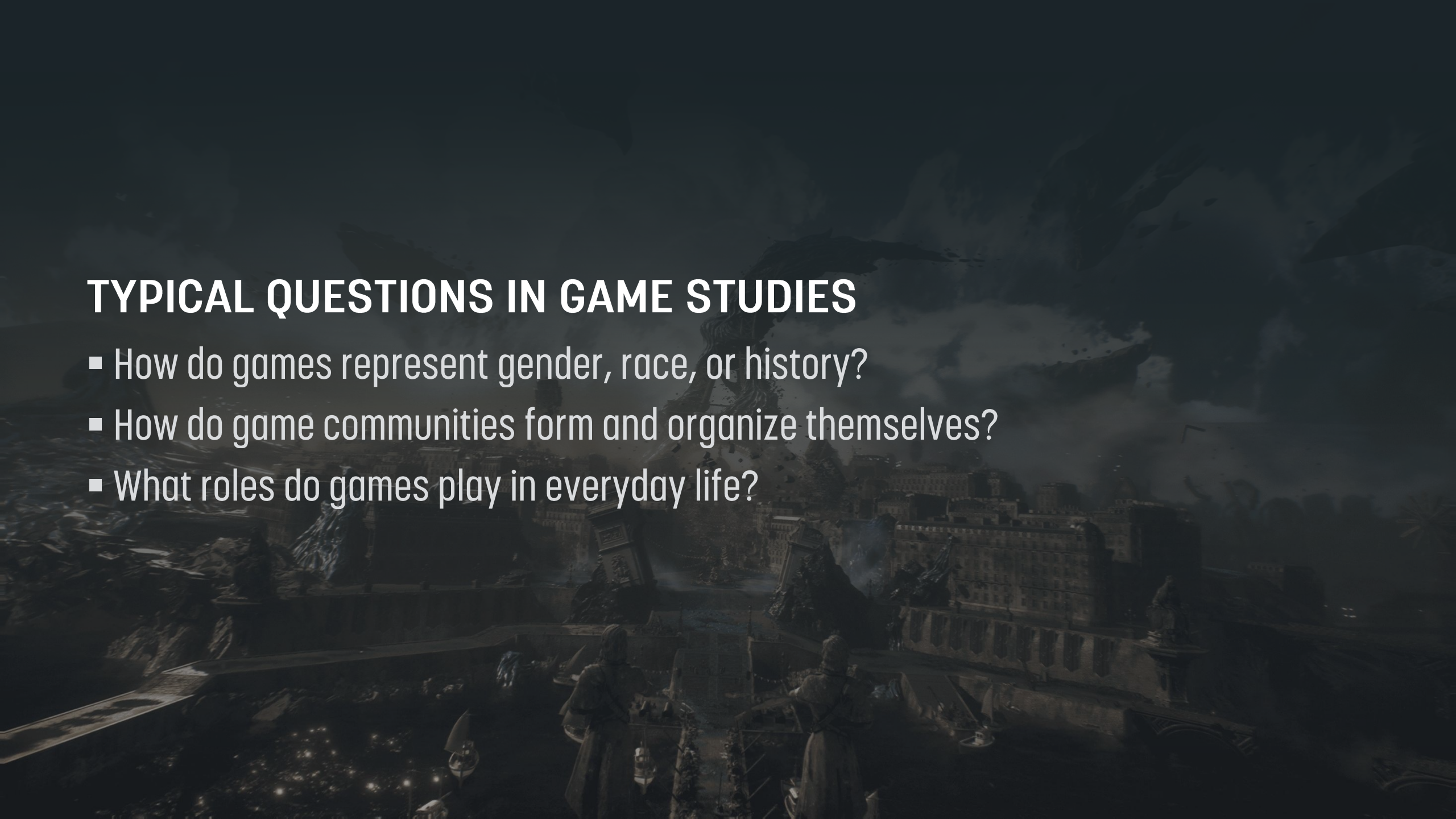
- **Gameplay and Mechanics:** Investigating how rules, goals, feedback systems, and difficulty curves affect player experience
- **Player Experience and Emotions:** Understanding how players feel while playing—immersion, flow state, frustration, excitement
- **Representation and Identity:** Exploring how gender, race, sexuality, and other identities are represented in games, and how these representations affect players' perceptions and experiences.

KEY TOPICS OF INQUIRY

- **Community and Culture:** Studying fan cultures, modding (game modifications), eSports, live-streaming communities, and the ways games foster social interaction
- **Serious Games and Gamification:** Investigating how game design principles can be applied to education, health, organizational management, and other non-entertainment contexts

TYPICAL QUESTIONS IN GAME STUDIES

- How do games represent gender, race, or history?
- How do game communities form and organize themselves?
- What roles do games play in everyday life?



GAMES AS CULTURAL ARTIFACTS

- How can games generate meaning through their formal elements (i.e., rules, mechanics, and aesthetics)
- Roots in humanities: games are "texts" to be decoded.
- Example: Undertale as critique of traditional hero narratives

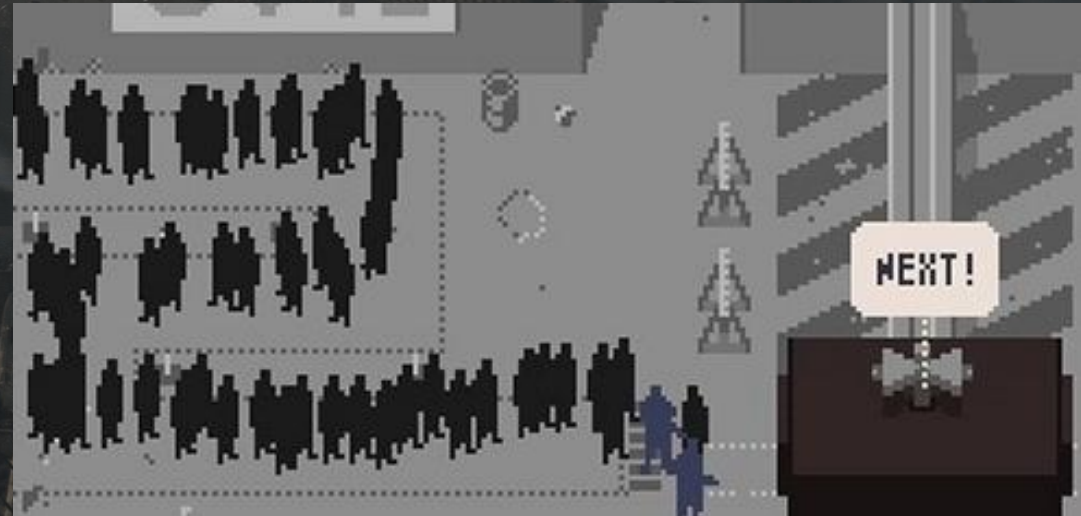


PLAYER PSYCHOLOGY AND BEHAVIORAL IMPACT

- Cognitive and Emotional Engagement
- Social science: empirical methods to study player motivations, emotions, and cognitive processes
- Player Experience Inventory (PXI) quantify immersion and flow states
- Ethnographic studies reveal how multiplayer environments foster collaboration, competition, and toxicity

3. GAME DESIGN AND TECHNOLOGY

- Procedural Rhetoric and Mechanics
- How game mechanics convey ideological messages
- Ian Bogost's concept of procedural rhetoric
- Example: Papers, Please critiques immigration policies through bureaucratic gameplay



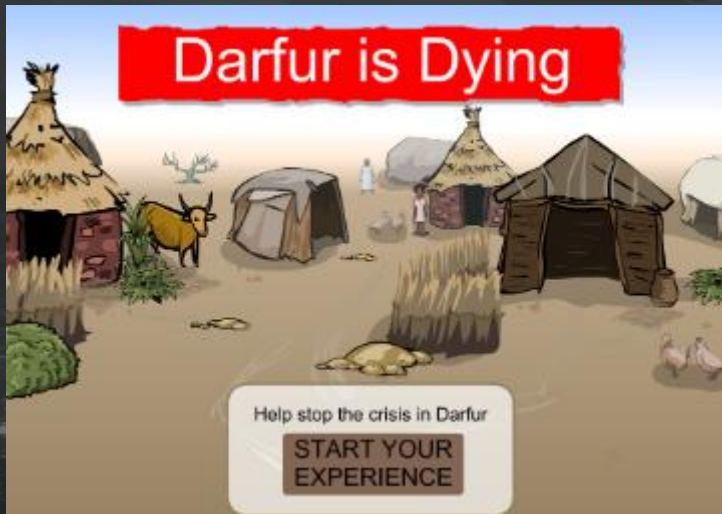
3. GAME DESIGN AND TECHNOLOGY

- Emerging Technologies
- AI tools like MidJourney
- Challenge notions of authorship



4. GAMES BEYOND ENTERTAINMENT

■ Serious Games and Gamification



KEY SCHOLARS AND INFLUENTIAL FIGURES

- **Espen Aarseth:** popularizing the term ludology & ergodic literature
- **Jesper Juul:** ludology vs. narratology discussion
- **Katie Salen and Eric Zimmerman:** theory and practice in game design
- **Ian Bogost:** procedural rhetoric
- **T.L. Taylor:** on online gaming communities
- **Mary Flanagan:** critical play and values in game design
- **Nick Yee:** player motivations and demographics
- **Janet Murray:** digital interactive narrative

EXAMPLE TOPIC: VIOLENCE IN GAMES

- Public debates about violent games and aggression
- Research explores complex links, not simple cause and effect
- Focus on context, player interpretation, and design



EXAMPLE TOPIC: HISTORICAL GAME STUDIES

- How games represent the past and historical events
- Accuracy vs. fun and playability
- Games as a way of learning about history



GAME STUDIES JOURNALS AND VENUES

- Game Studies (online, open access)
- Special issues in media and communication journals
- Conferences focusing on games and culture



The screenshot shows the homepage of the Game Studies journal. The header features the title "Game Studies" in a pixelated font, with the subtitle "the international journal of computer game research" and navigation links for "home", "about", and "archive". The main content area includes a welcome message for the first issue, "Computer Game Studies, Year One", by Espen Aarseth. It also lists two featured articles: "Beyond Myth and Metaphor: The Case of Narrative in Digital Media" by Marie-Laure Ryan and "Games telling stories?" by Jesper Juul.

Game Studies the international journal of computer game research

home about archive

Computer Game Studies, Year One
by Espen Aarseth

Welcome to the first issue of the first academic, peer-reviewed journal dedicated to computer game studies. This is a noteworthy occasion, and perhaps the most remarkable aspect is that such a journal has not been started before. As we know, there have been computer games for almost as long as there have been computers: *SpaceWar*, arguably the first *modern* game, turns forty this year, and commercially the genre has existed for three decades. So why not something like this before?
[\[more\]](#)

Beyond Myth and Metaphor: The Case of Narrative in Digital Media
by Marie-Laure Ryan

If we compare the field of digital textuality to other areas of study in the humanities, its most striking feature is the precedence of theory over the object of study. Most of us read novels and see movies before

Games telling stories?
by Jesper Juul

As questions go, this is not a bad one: Do games tell stories? Answering this should tell us both *how* to study games and *who* should study them. The affirmative answer suggests that games are easily studied from



WHAT GAME STUDIES IS NOT

- Not just game reviews or personal opinions
- Not purely about technical graphics or performance
- Goes beyond "is this game good or bad?"



WHY GAME STUDIES MATTERS FOR DESIGNERS

- Offers language to talk about representation and meaning
- Helps anticipate how different players might react
- Encourages critical reflection on design choices



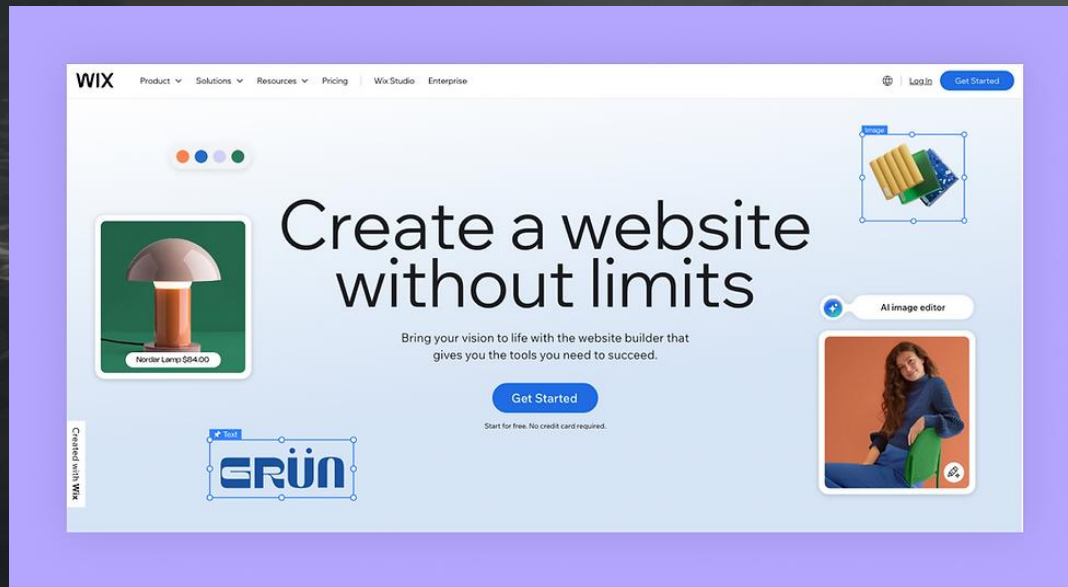
PART 2 – GAMES & HCI

- Human-Computer Interaction perspective on games
- How interaction shapes player experience
- From usability to emotions and engagement



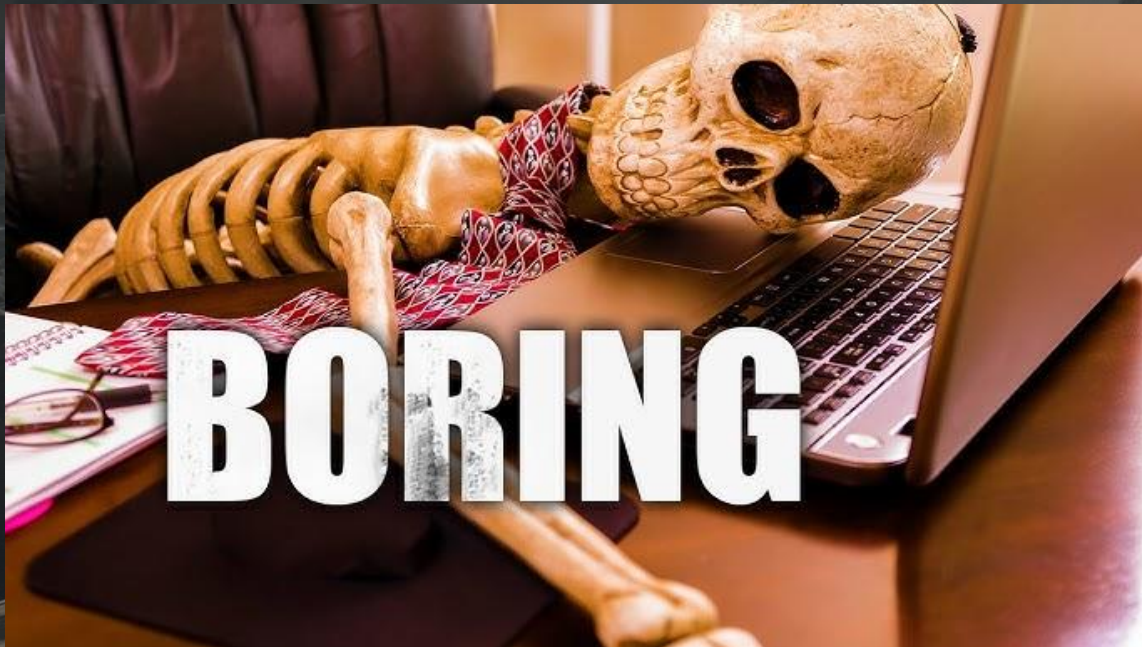
WHAT IS HCI?

- Field studying how people interact with technology
- Focus on usability, usefulness, and user experience
- Games are a special, playful case of HCI



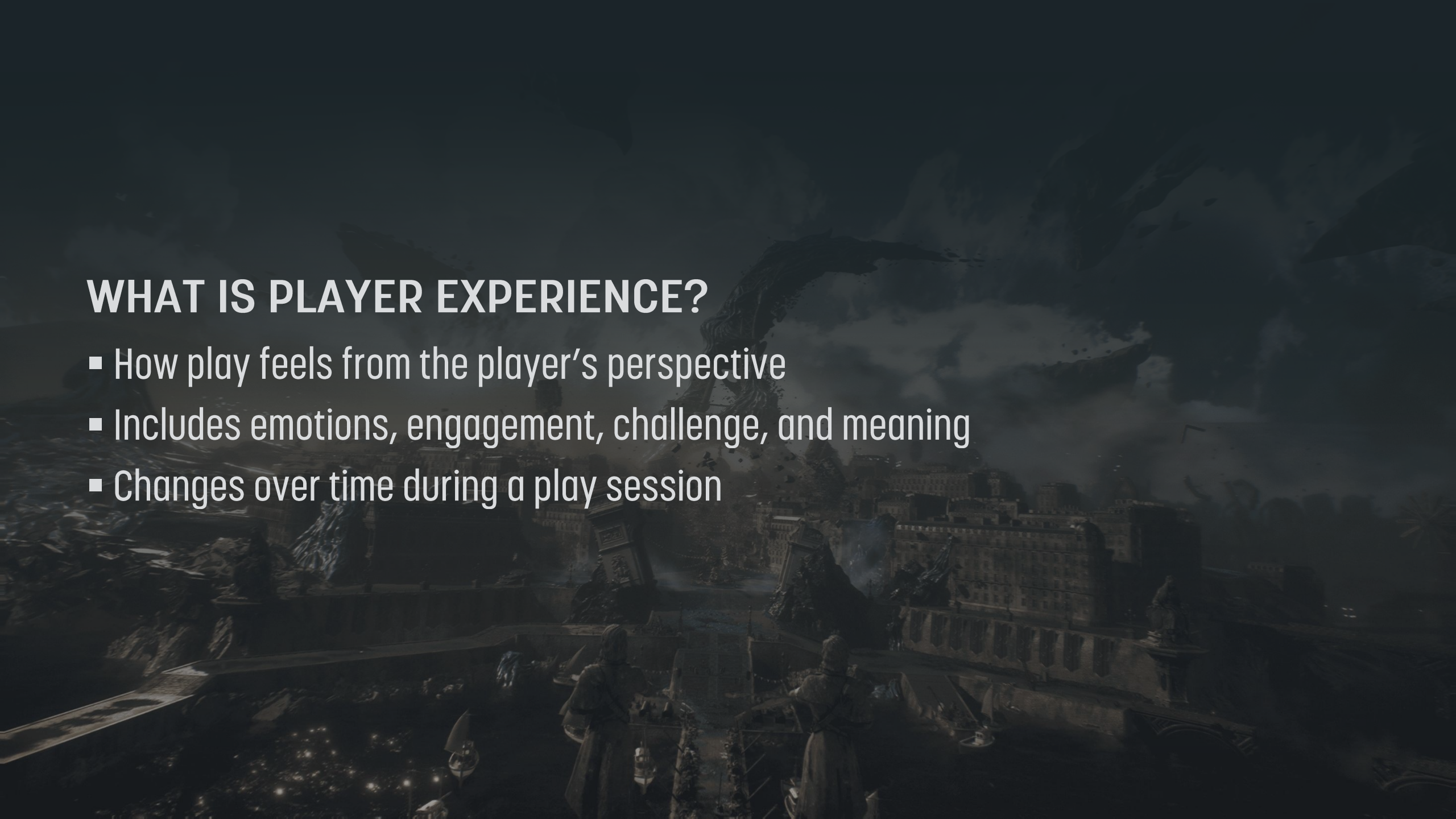
WHY GAMES ARE SPECIAL FOR HCI

- Goal is often fun, not efficiency
- Players accept challenge and even frustration
- Design must balance usability with meaningful difficulty



WHAT IS PLAYER EXPERIENCE?

- How play feels from the player's perspective
- Includes emotions, engagement, challenge, and meaning
- Changes over time during a play session



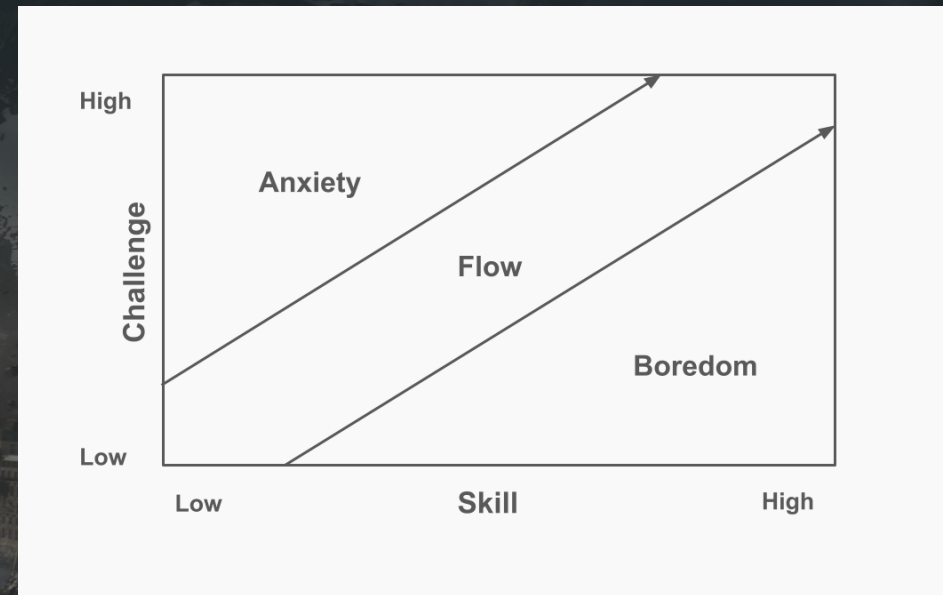
DIMENSIONS OF PLAYER EXPERIENCE

- Flow: intense focus and smooth challenge
- Immersion and presence: feeling inside the game world
- Competence, autonomy, relatedness, and emotion



FLOW IN GAMES

- Balance between challenge and skill
- Clear goals and immediate feedback
- Losing track of time while playing



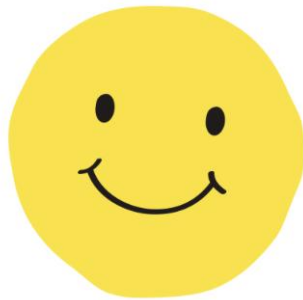
IMMERSION AND PRESENCE

- Sense of being "inside" the game world
- Can come from visuals, sound, story, or mechanics
- Not only about realism; stylized games can be immersive too



EMOTIONS IN GAMES

- Moment-to-moment emotions: joy, fear, tension, relief
- Long-term feelings: attachment to characters or worlds
- Mixed emotions: frustration and satisfaction intertwined



Joy



Sadness



Anger



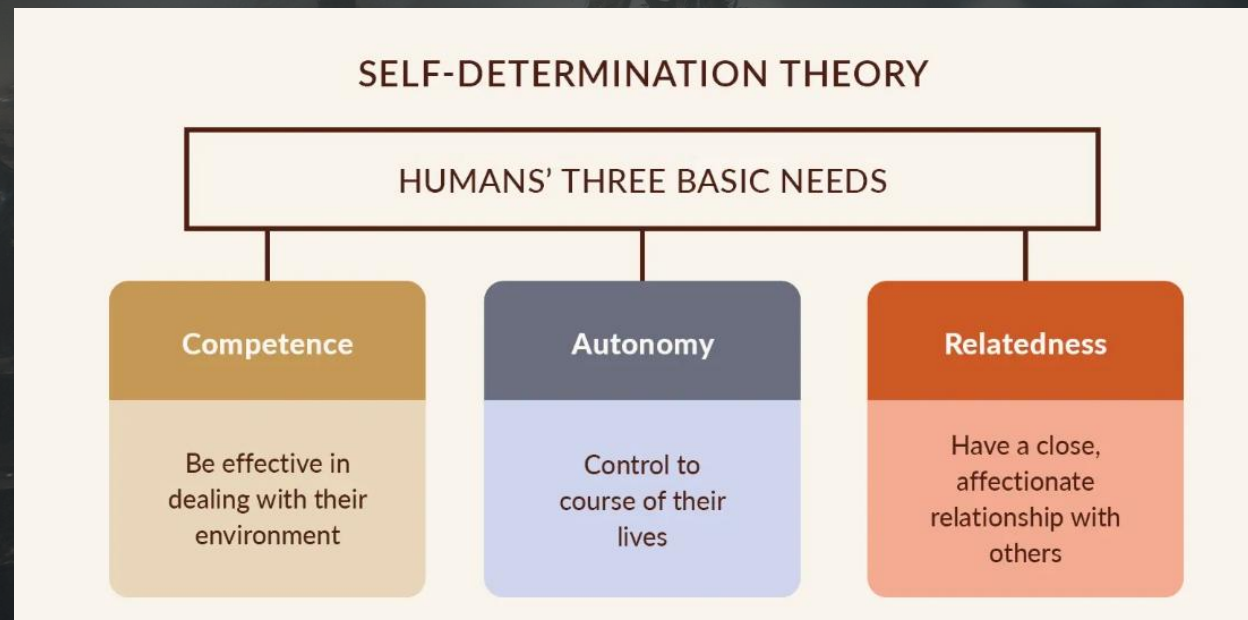
Fear



Disgust

MOTIVATION AND SELF-DETERMINATION

- Competence: feeling capable and effective
- Autonomy: feeling of choice and control
- Relatedness: feeling connected to others



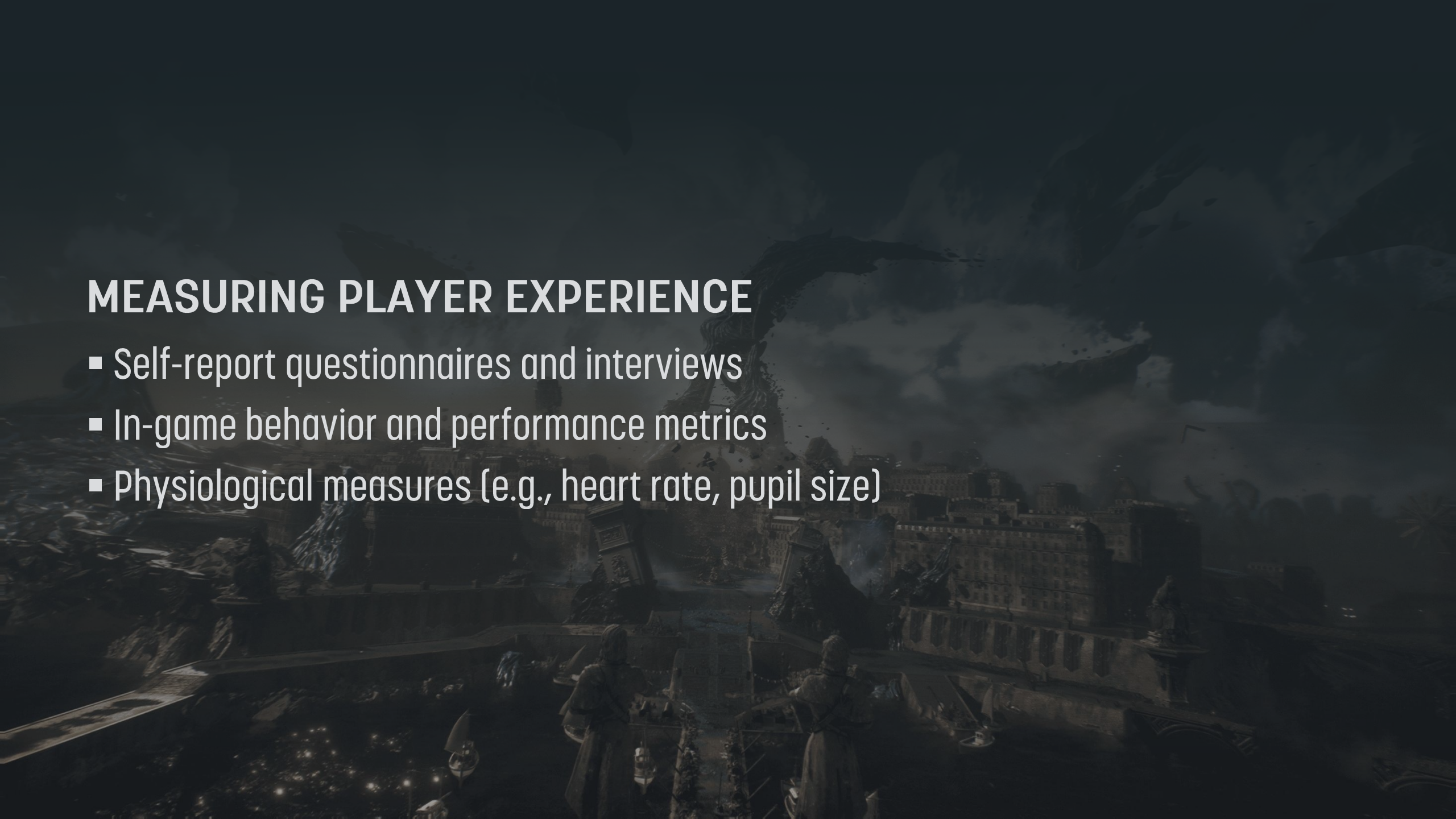
SOCIAL PLAY AND CONNECTEDNESS

- Games as shared experiences with friends or strangers
- Co-op, competitive play, and spectatorship
- Benefits and challenges: community vs. toxicity



MEASURING PLAYER EXPERIENCE

- Self-report questionnaires and interviews
- In-game behavior and performance metrics
- Physiological measures (e.g., heart rate, pupil size)



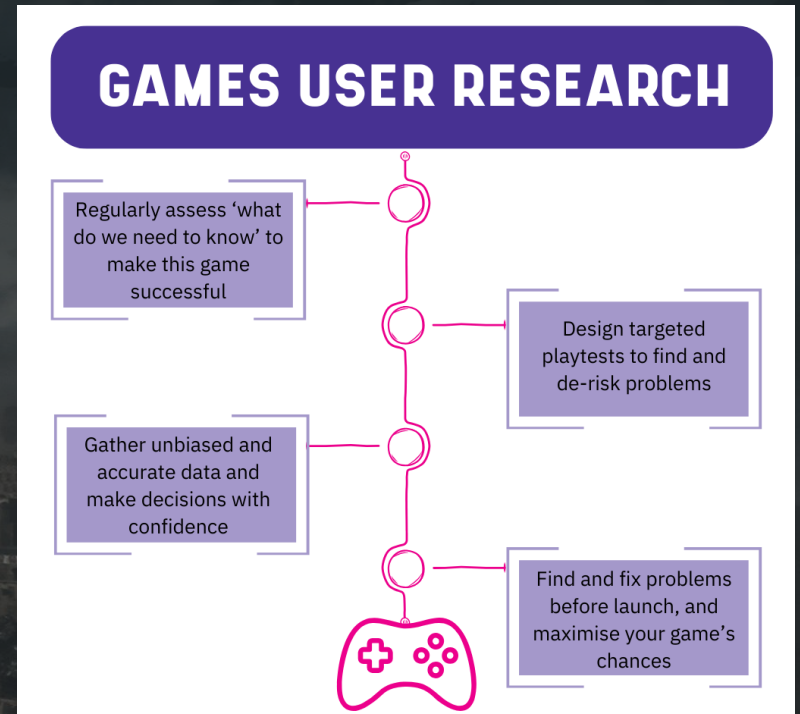
WHY PLAYER EXPERIENCE MATTERS FOR YOU

- Guides design decisions beyond "I like it"
- Helps you design for diverse players
- Supports communication in teams and with stakeholders



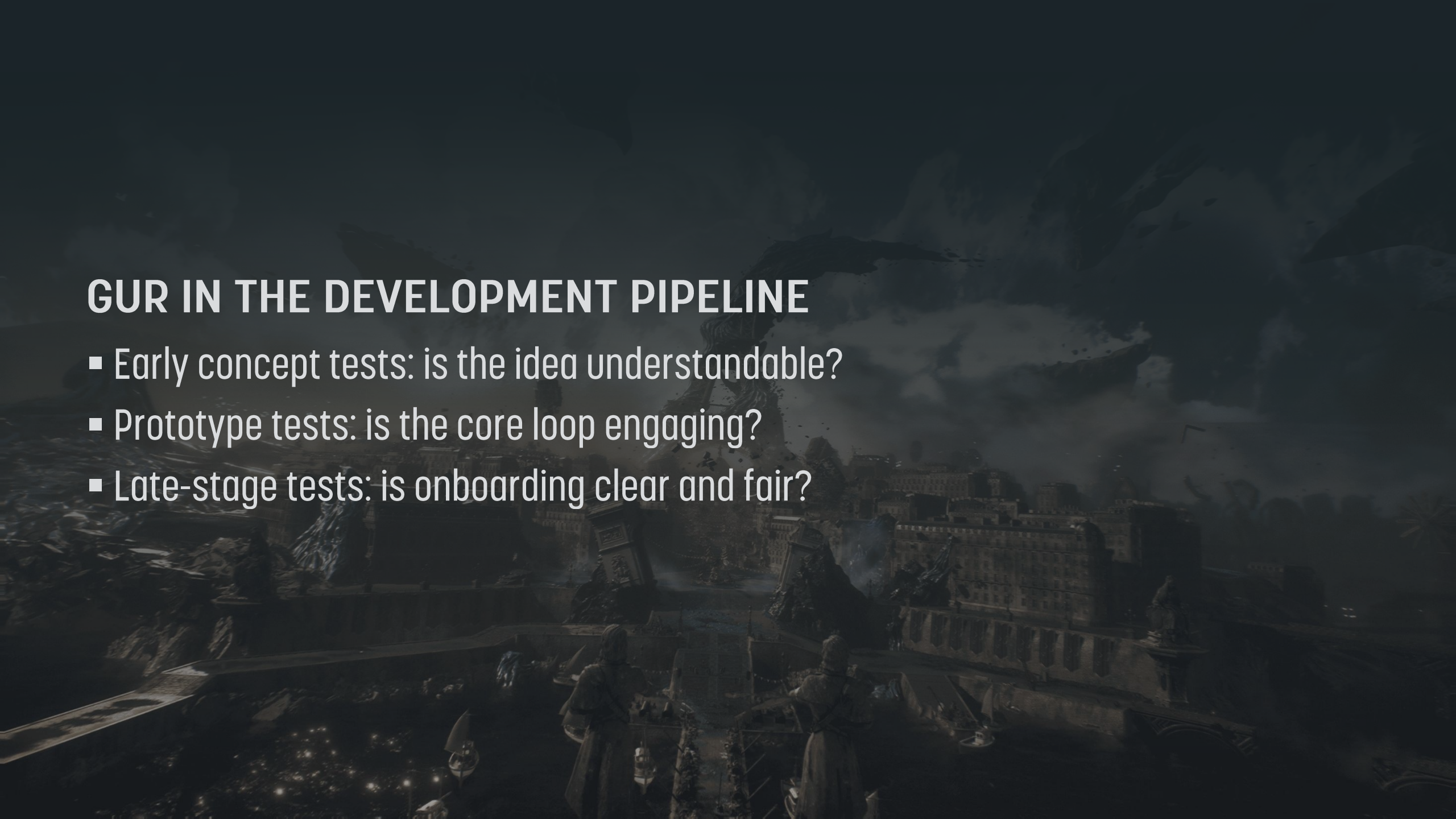
PART 3 – GAMES USER RESEARCH (GUR)

- Research embedded in game development
- Focus on specific games and design questions
- Bridges design, analytics, and player feedback



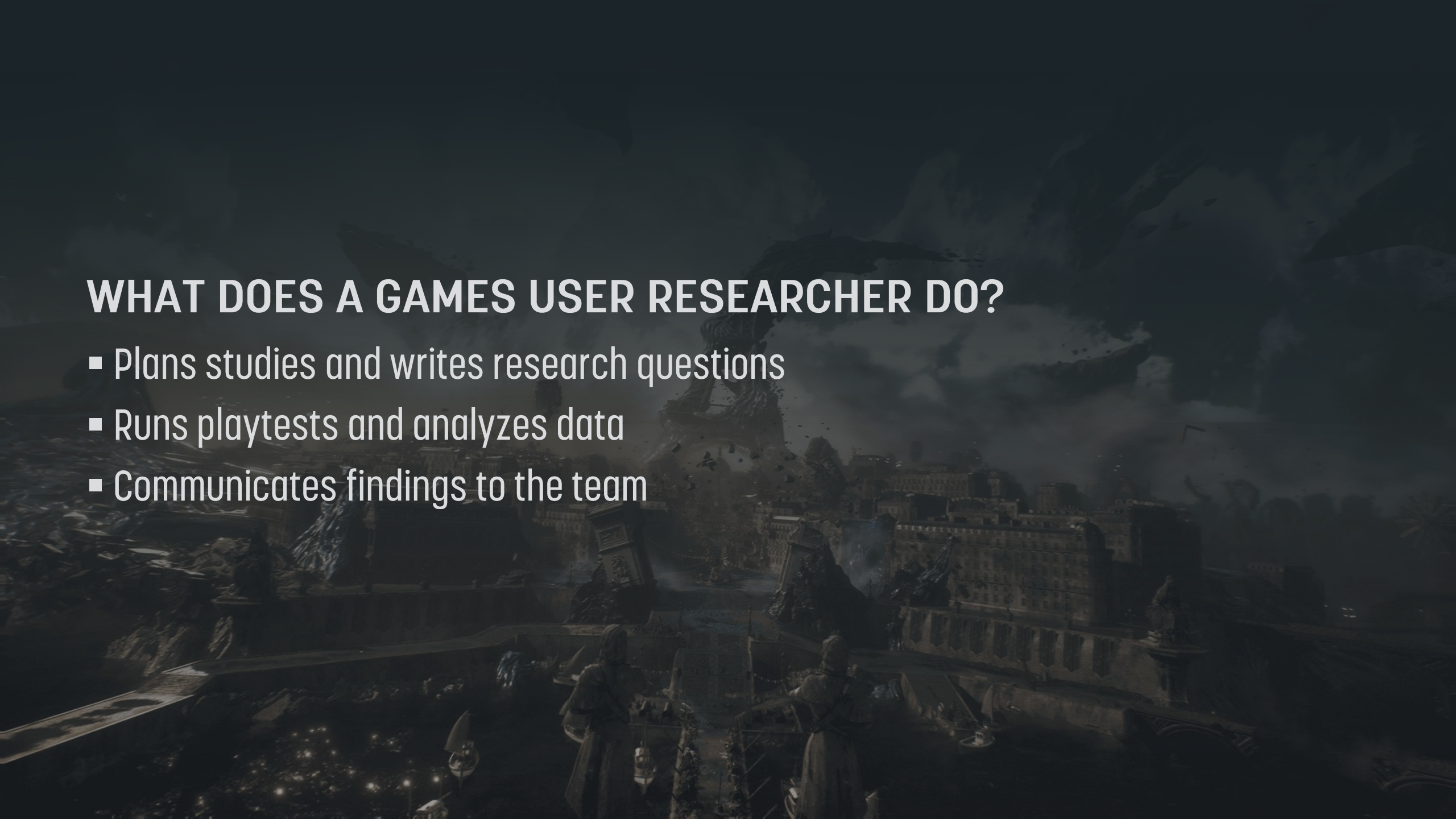
GUR IN THE DEVELOPMENT PIPELINE

- Early concept tests: is the idea understandable?
- Prototype tests: is the core loop engaging?
- Late-stage tests: is onboarding clear and fair?



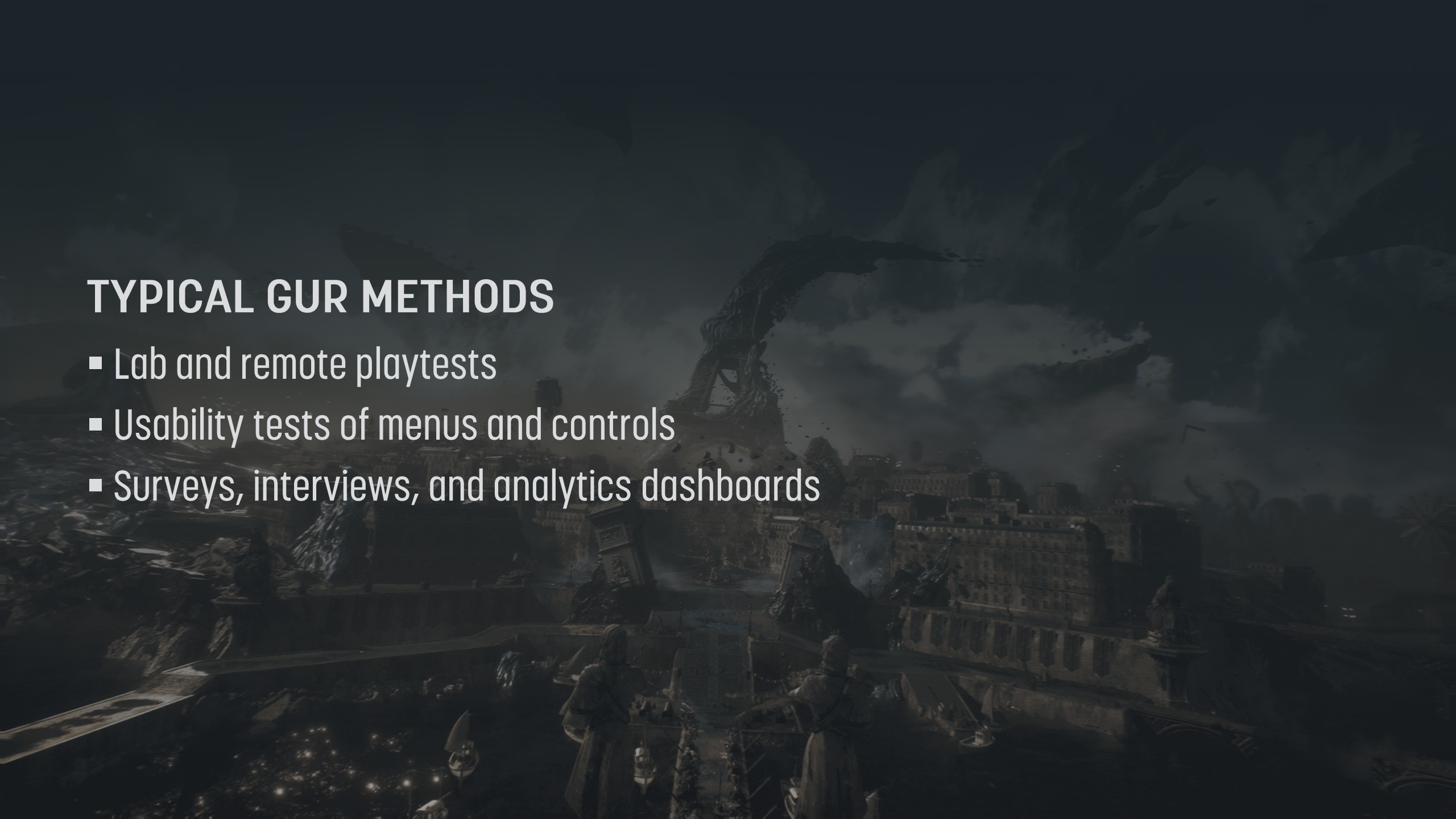
WHAT DOES A GAMES USER RESEARCHER DO?

- Plans studies and writes research questions
- Runs playtests and analyzes data
- Communicates findings to the team



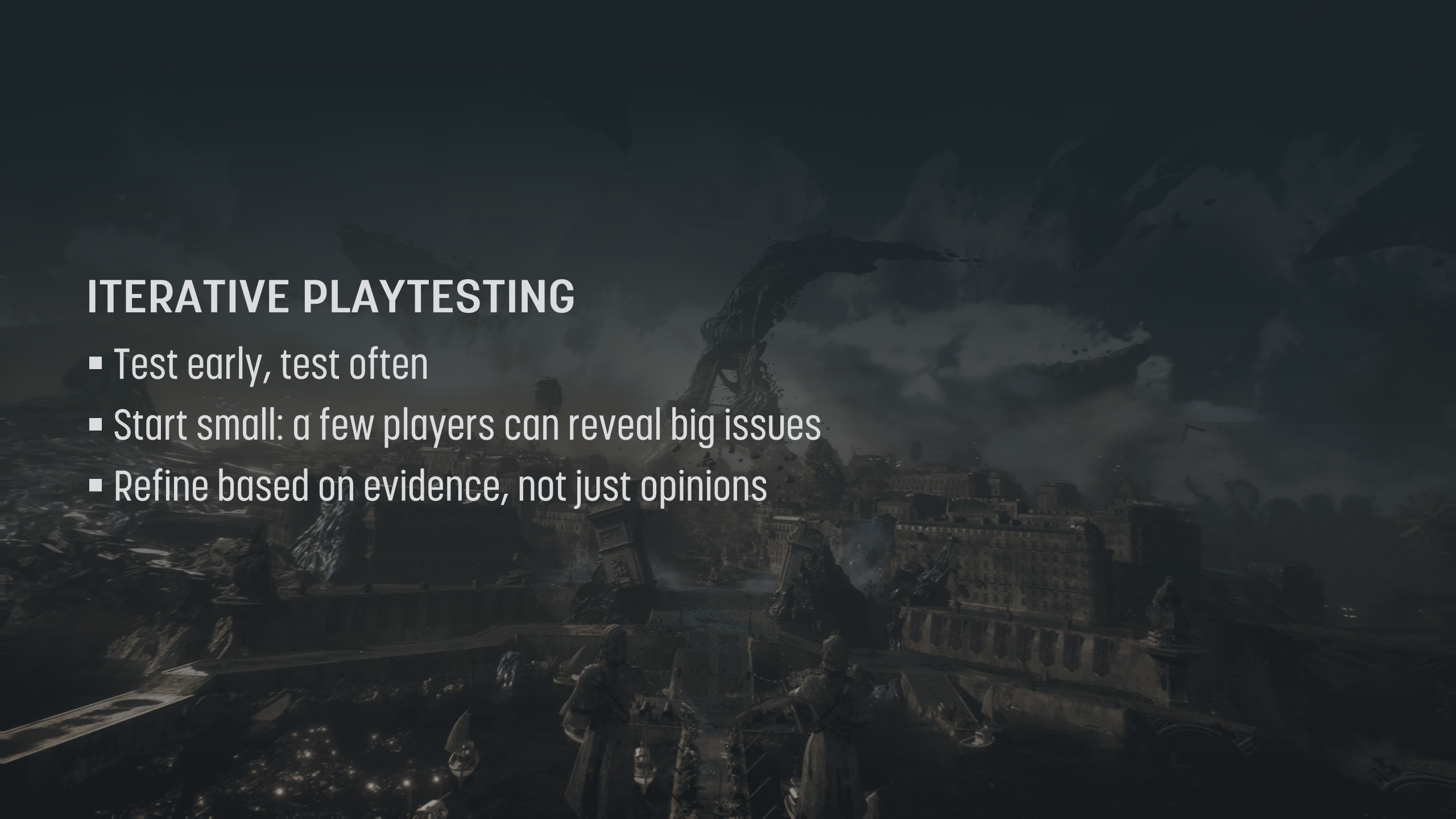
TYPICAL GUR METHODS

- Lab and remote playtests
- Usability tests of menus and controls
- Surveys, interviews, and analytics dashboards



ITERATIVE PLAYTESTING

- Test early, test often
- Start small: a few players can reveal big issues
- Refine based on evidence, not just opinions



CASE EXAMPLE: ONBOARDING AND TUTORIALS

- Observe where players get stuck in the first 10-15 minutes
- Check if goals and controls are clear
- Use findings to adjust pacing, hints, and feedback



CASE EXAMPLE: BALANCING DIFFICULTY

- Compare performance and frustration across different settings
- Collect self-reports of challenge and enjoyment
- Aim for a satisfying, not trivial, difficulty curve



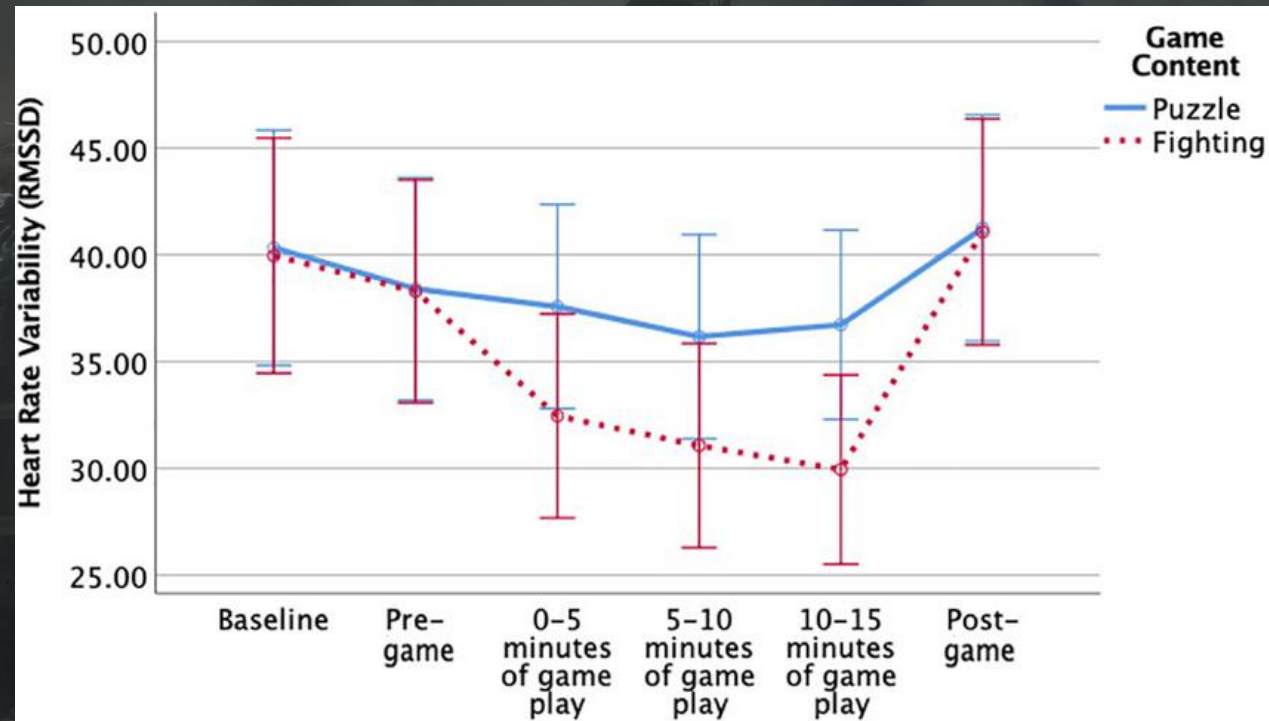
USING TELEMETRY AND ANALYTICS

- Automatic logging of in-game events and behavior
- Heatmaps of deaths, paths, and interactions
- A/B tests to compare design variants



BIOMETRICS IN GAMES RESEARCH

- Heart rate, skin conductance, eye tracking, and more
- Can reveal arousal and attention patterns
- Require careful interpretation and ethical handling



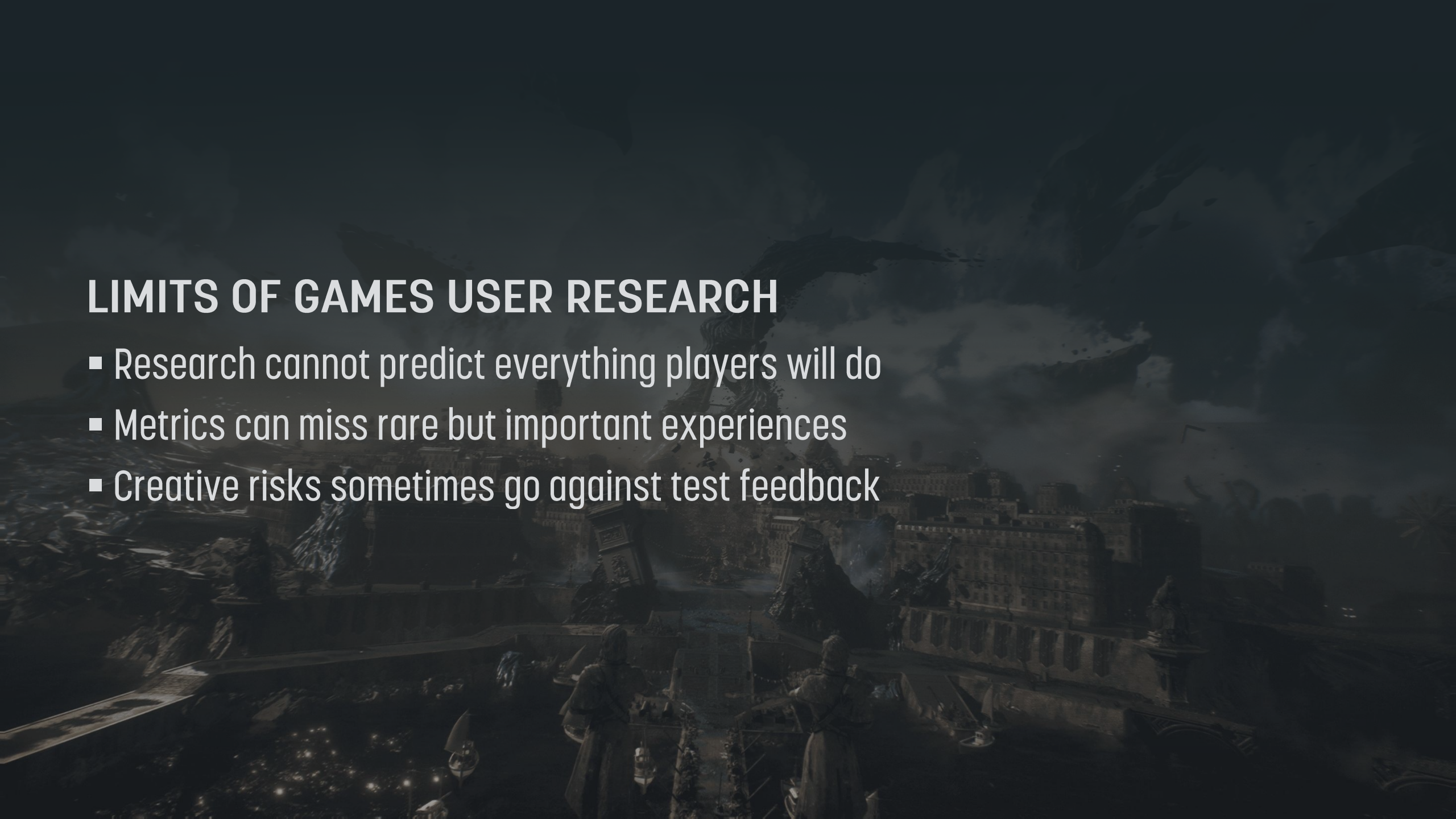
ETHICS AND TRANSPARENCY IN GAME DATA

- Players should know what is measured and why
- Respect privacy and data protection laws
- Avoid manipulative or exploitative designs



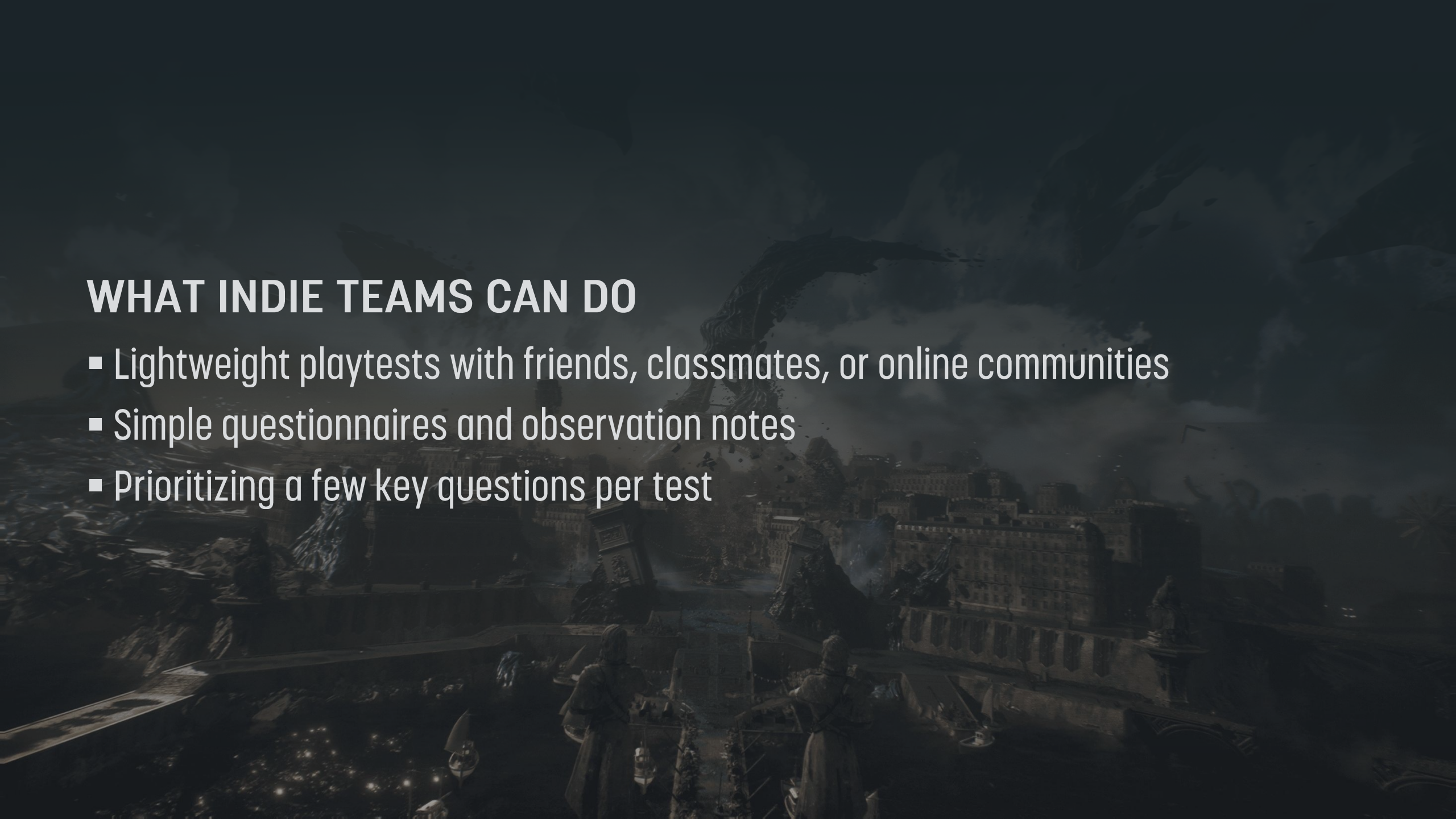
LIMITS OF GAMES USER RESEARCH

- Research cannot predict everything players will do
- Metrics can miss rare but important experiences
- Creative risks sometimes go against test feedback



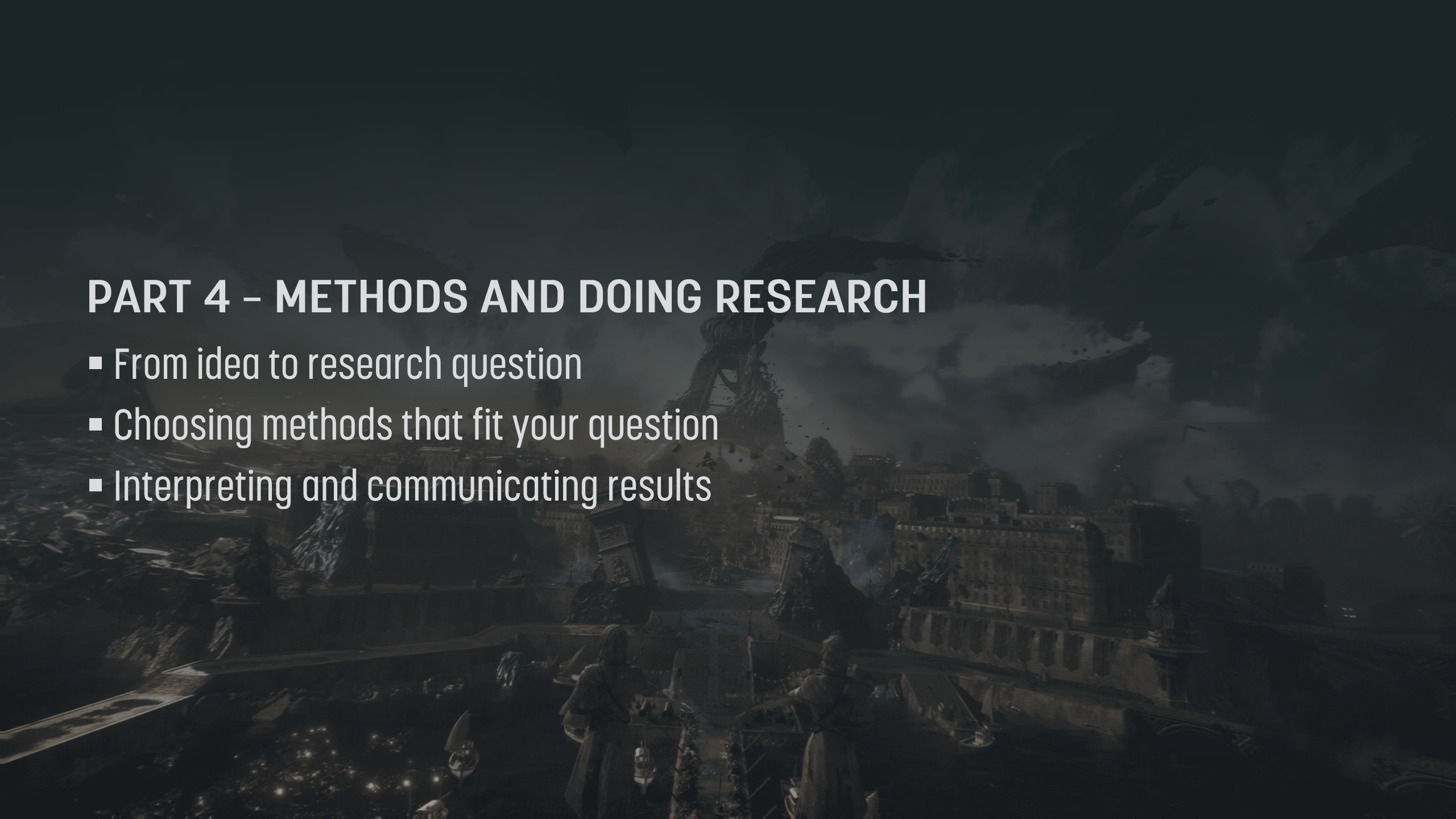
WHAT INDIE TEAMS CAN DO

- Lightweight playtests with friends, classmates, or online communities
- Simple questionnaires and observation notes
- Prioritizing a few key questions per test



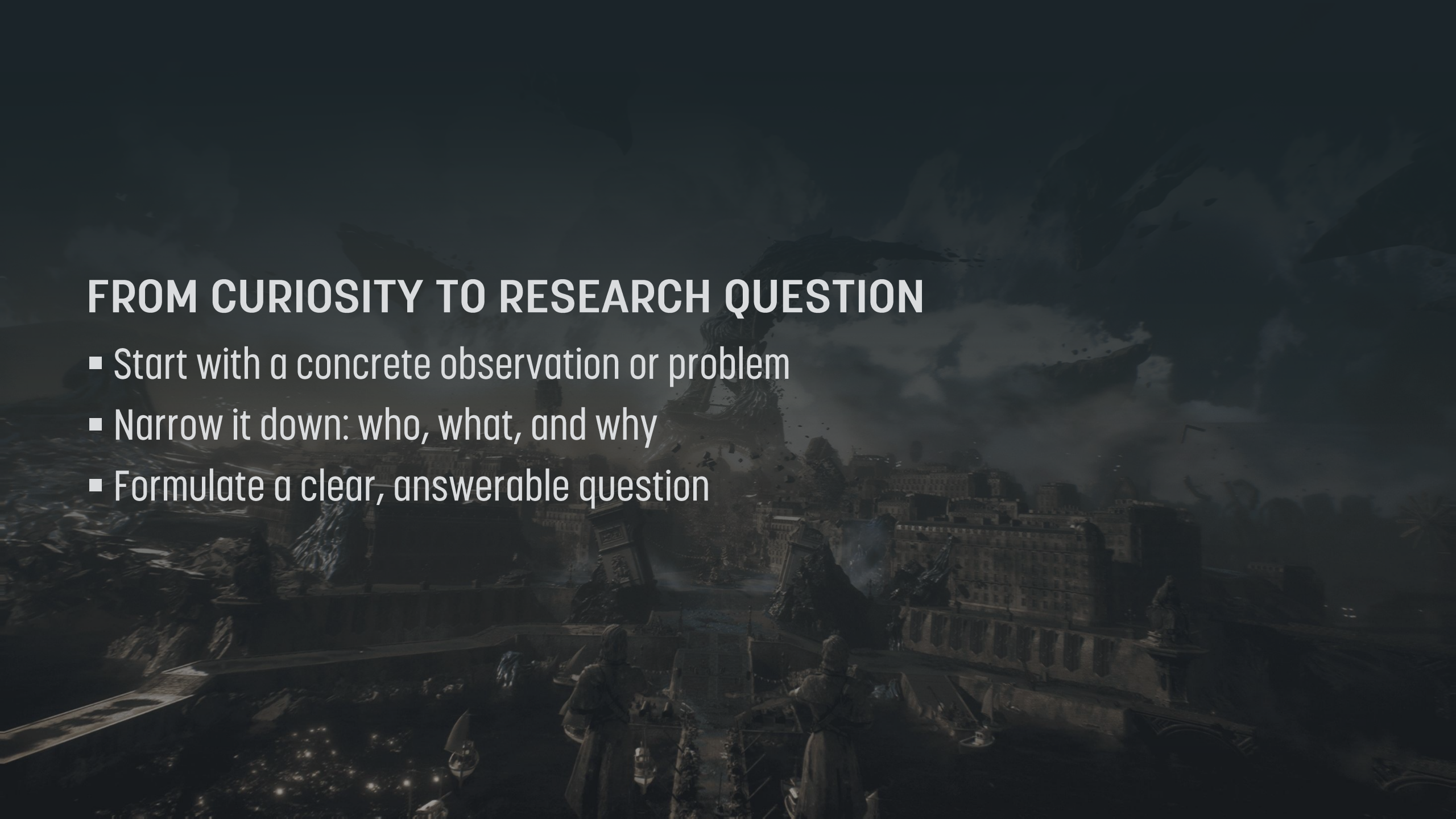
PART 4 – METHODS AND DOING RESEARCH

- From idea to research question
- Choosing methods that fit your question
- Interpreting and communicating results



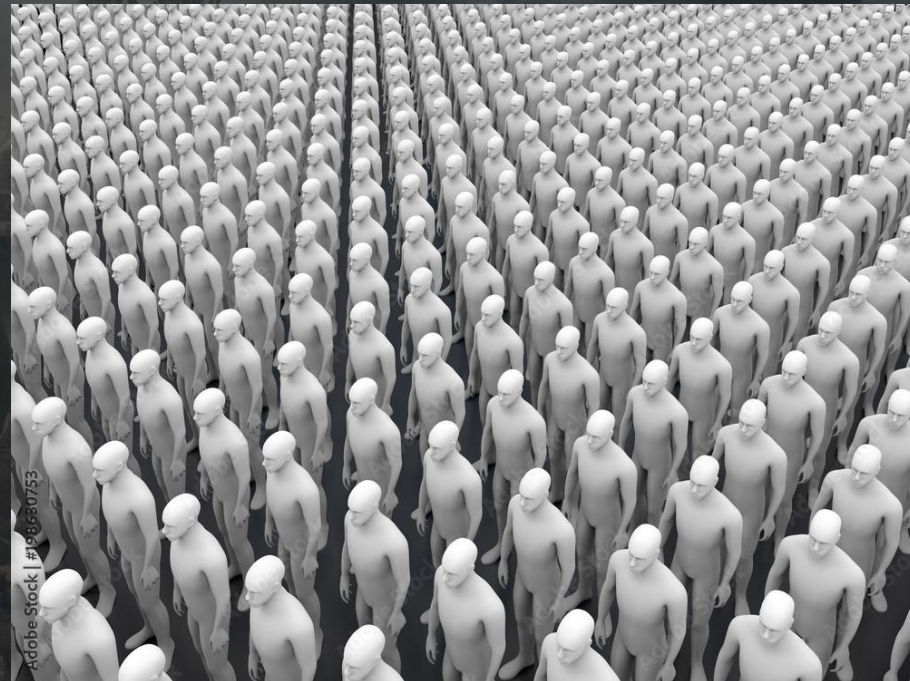
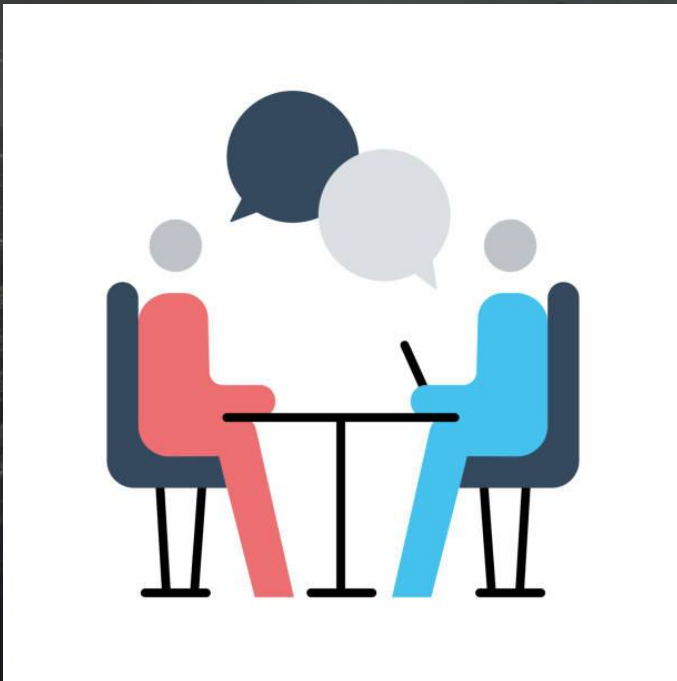
FROM CURIOSITY TO RESEARCH QUESTION

- Start with a concrete observation or problem
- Narrow it down: who, what, and why
- Formulate a clear, answerable question



QUANTITATIVE VS. QUALITATIVE APPROACHES

- Quantitative: numbers, patterns, comparisons
- Qualitative: stories, meanings, and interpretations
- Often combined in mixed-methods studies



COMMON QUANTITATIVE STUDY DESIGNS

- Experiments comparing different versions of a game
- Surveys with rating scales and large samples
- Analytics-based analyses of in-game behavior



COMMON QUALITATIVE APPROACHES

- Interviews about players' experiences and motivations
- Observation of play sessions and think-aloud protocols
- Ethnographic studies of game communities



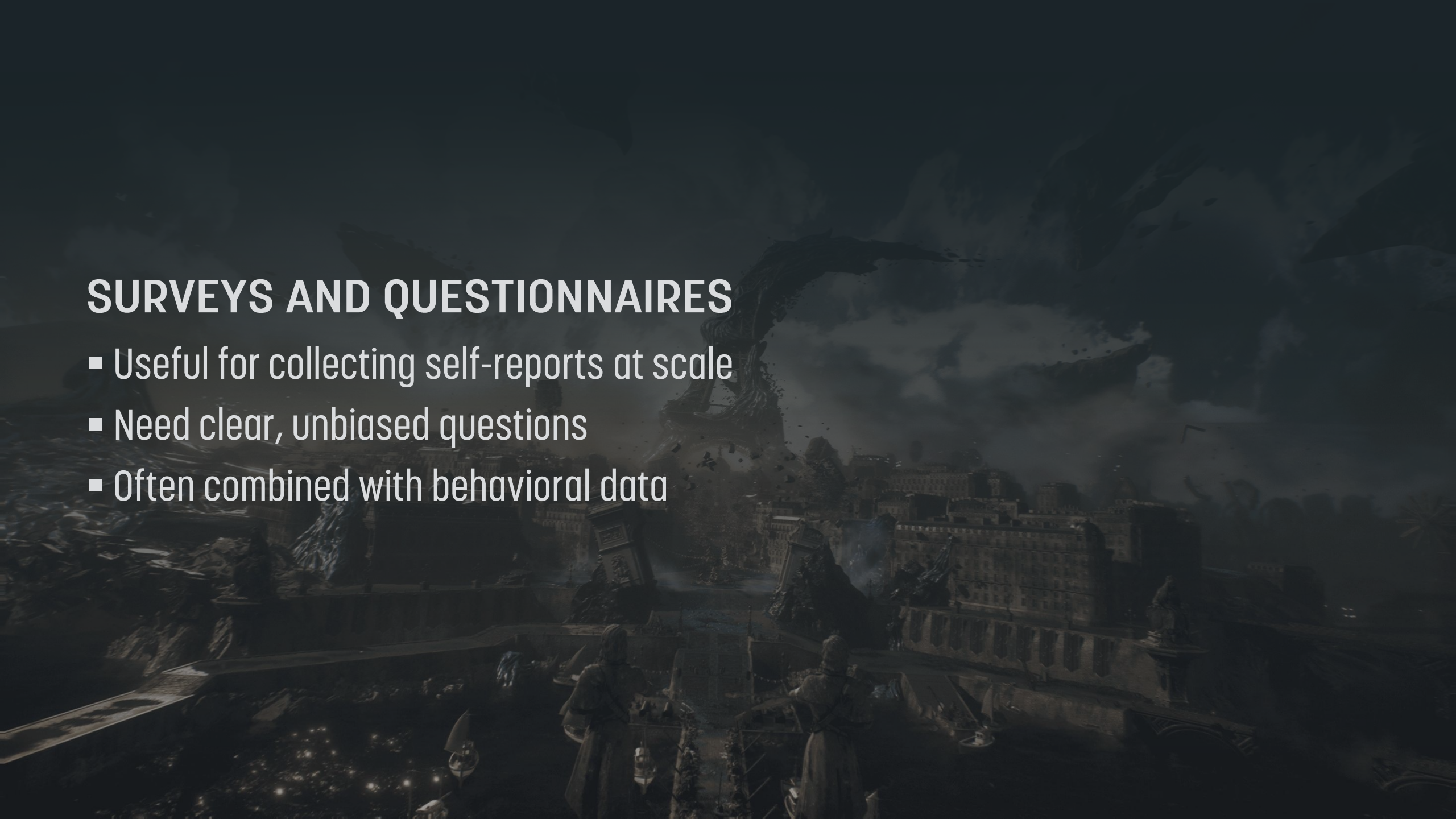
LAB STUDIES VS. FIELD STUDIES

- Lab: controlled environment, fewer confounding factors
- Field: natural context, more realism and messiness
- Each has strengths and weaknesses



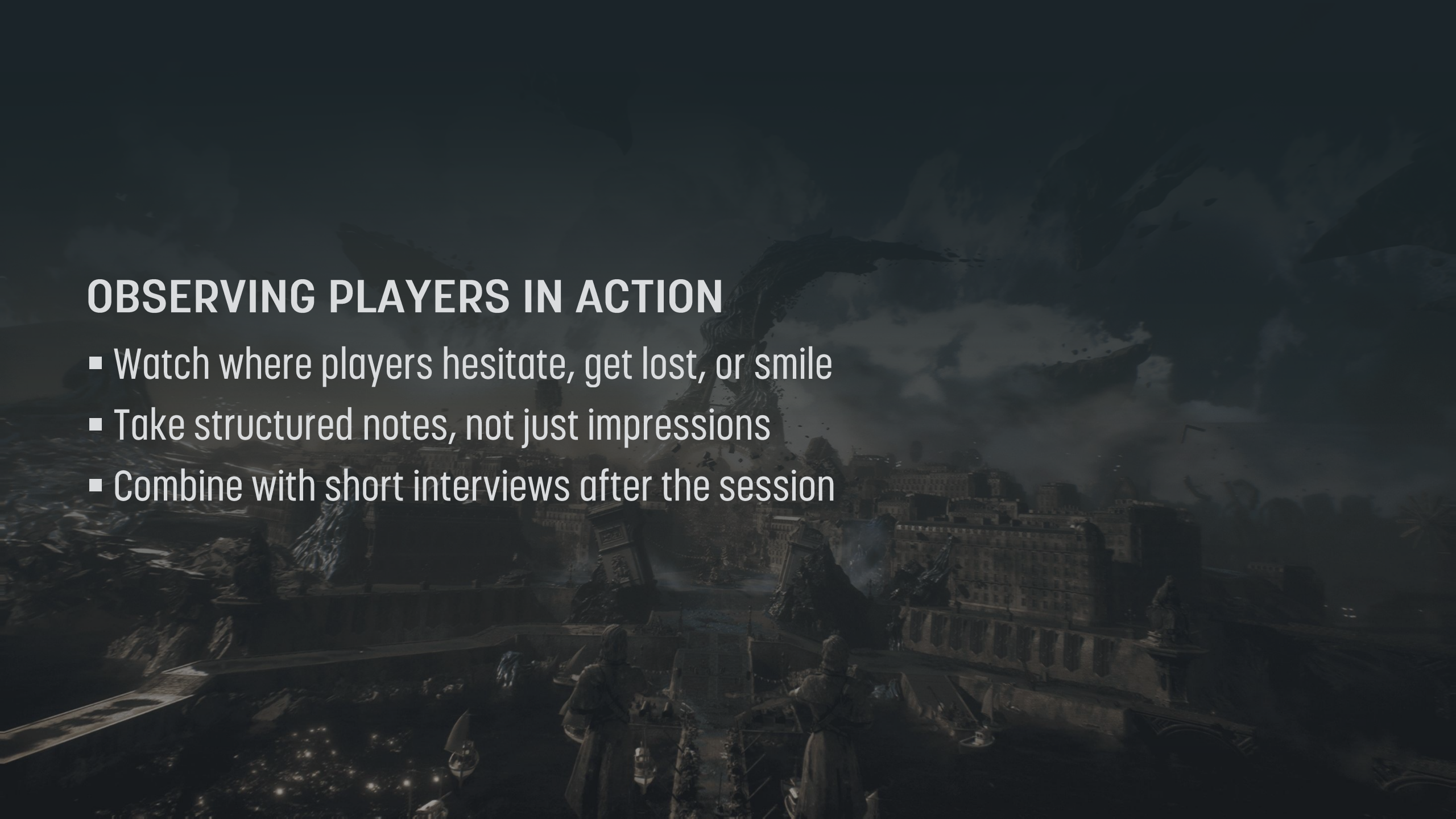
SURVEYS AND QUESTIONNAIRES

- Useful for collecting self-reports at scale
- Need clear, unbiased questions
- Often combined with behavioral data



OBSERVING PLAYERS IN ACTION

- Watch where players hesitate, get lost, or smile
- Take structured notes, not just impressions
- Combine with short interviews after the session



Michael Lankes, Günter Wallner, and Martin Kocur
CHI 2025, Yokohama, Japan

PATHWAYS OF DESIRE

ENHANCING NAVIGATION AND SENSE OF COMMUNITY THROUGH PLAYER-GENERATED
DESIRE PATHS



UNIVERSITY
OF APPLIED SCIENCES
UPPER AUSTRIA



Why Navigation Matters

- Navigation is central to many game genres
- poor navigation = lower enjoyment

MOTIVATION



The Concept of Desire Paths



Desire Paths in Commercial Games

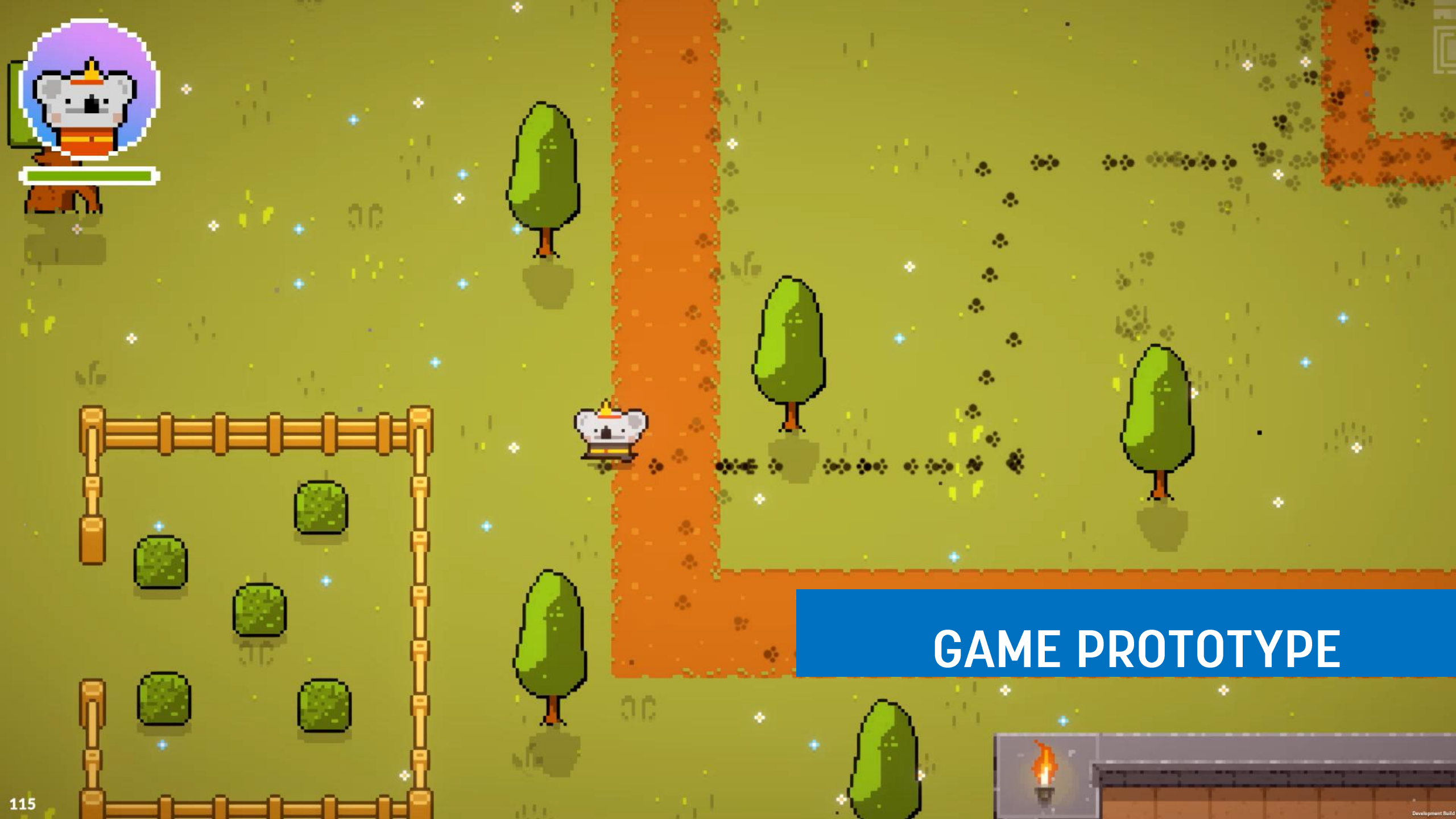


https://www.reddit.com/r/deathstranding/comments/wpy08w/desire_path_in_death_stranding/#lightbox

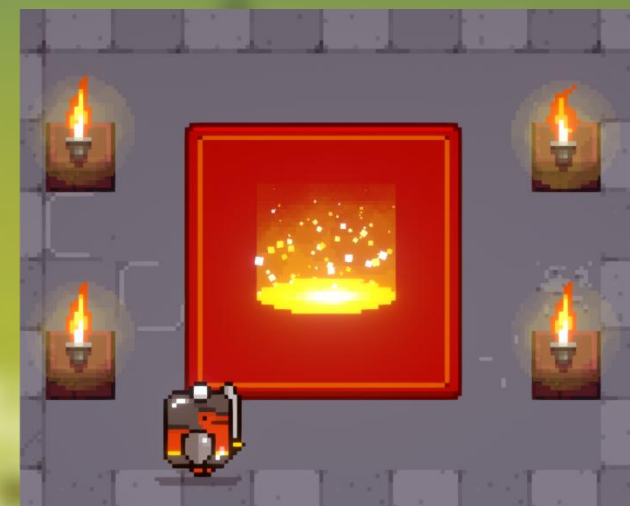
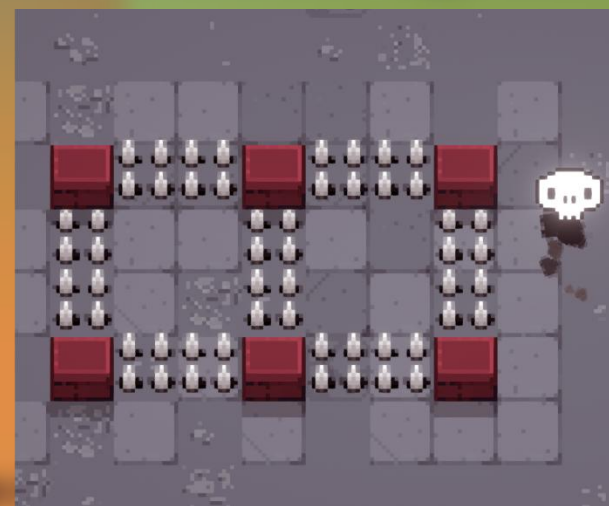
Research Focus

- Gameplay behavior
- Utility vs. Disturbance
- Sense of community
- Creative expression

QUESTION



GAME PROTOTYPE



LEVEL LAYOUT & OBJECTIVES

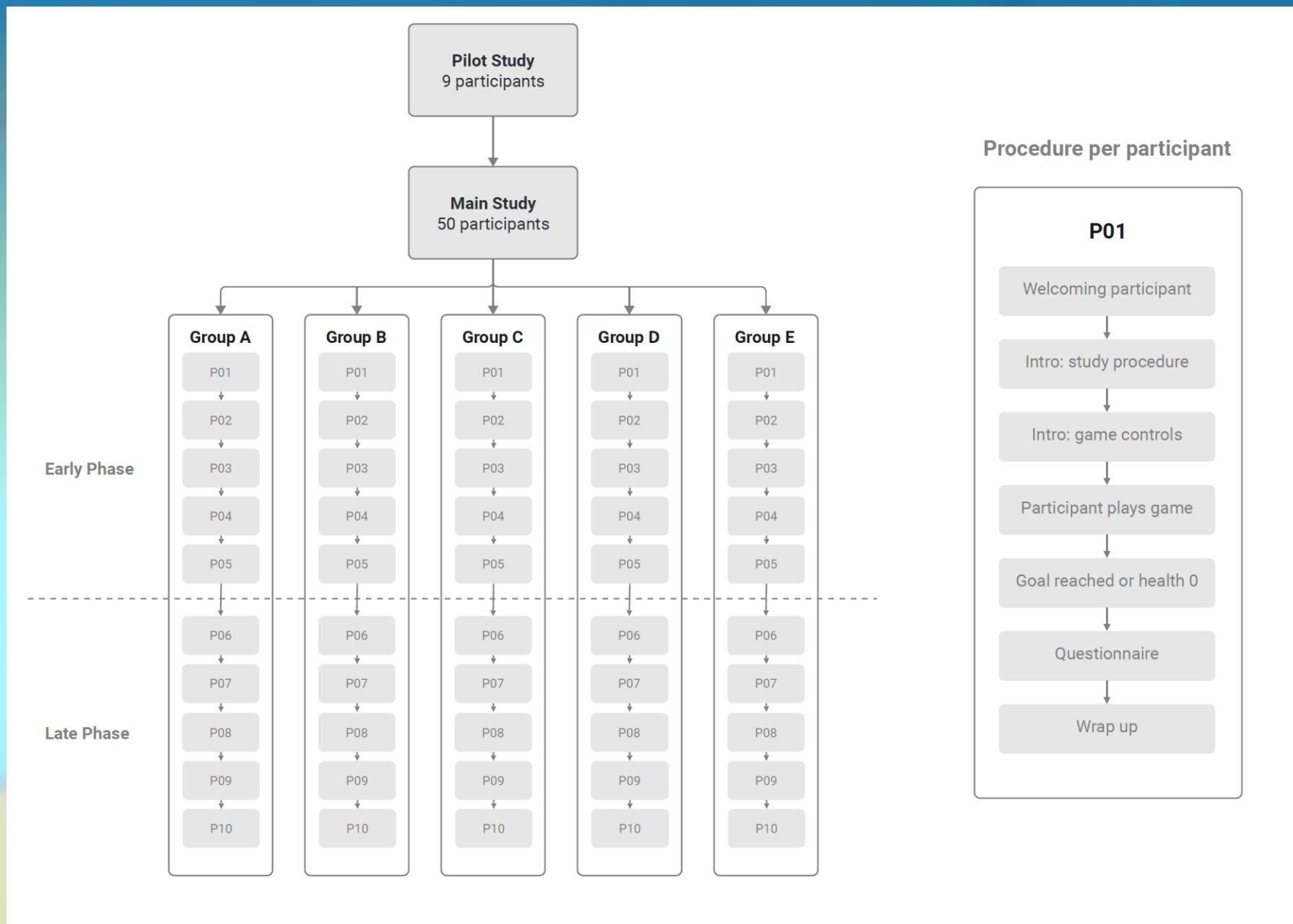


FOOTPRINT MECHANICS

Findings

- Teleporters actually confused players
- Footprints initially were all fully opaque

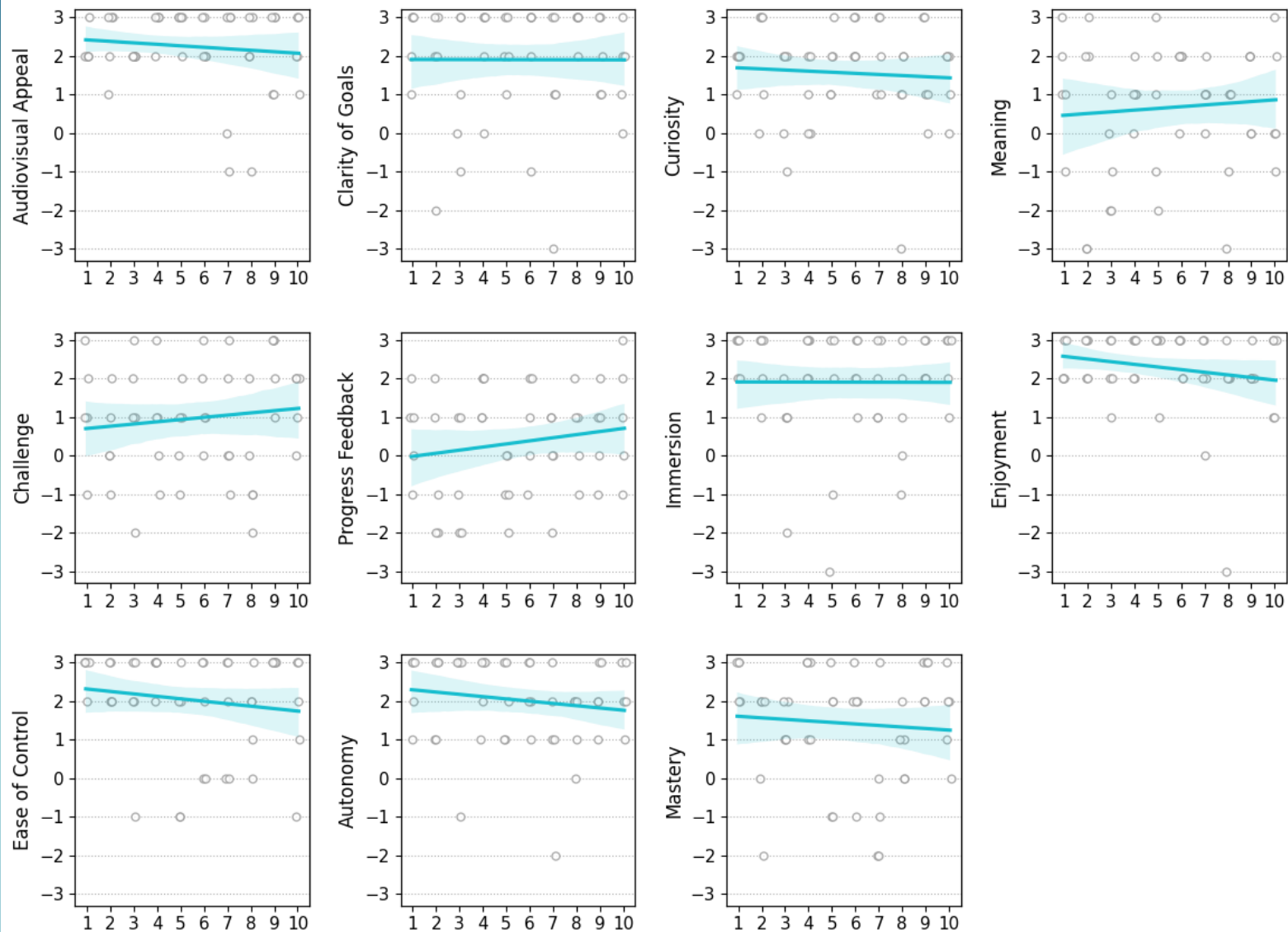
PILOT STUDY



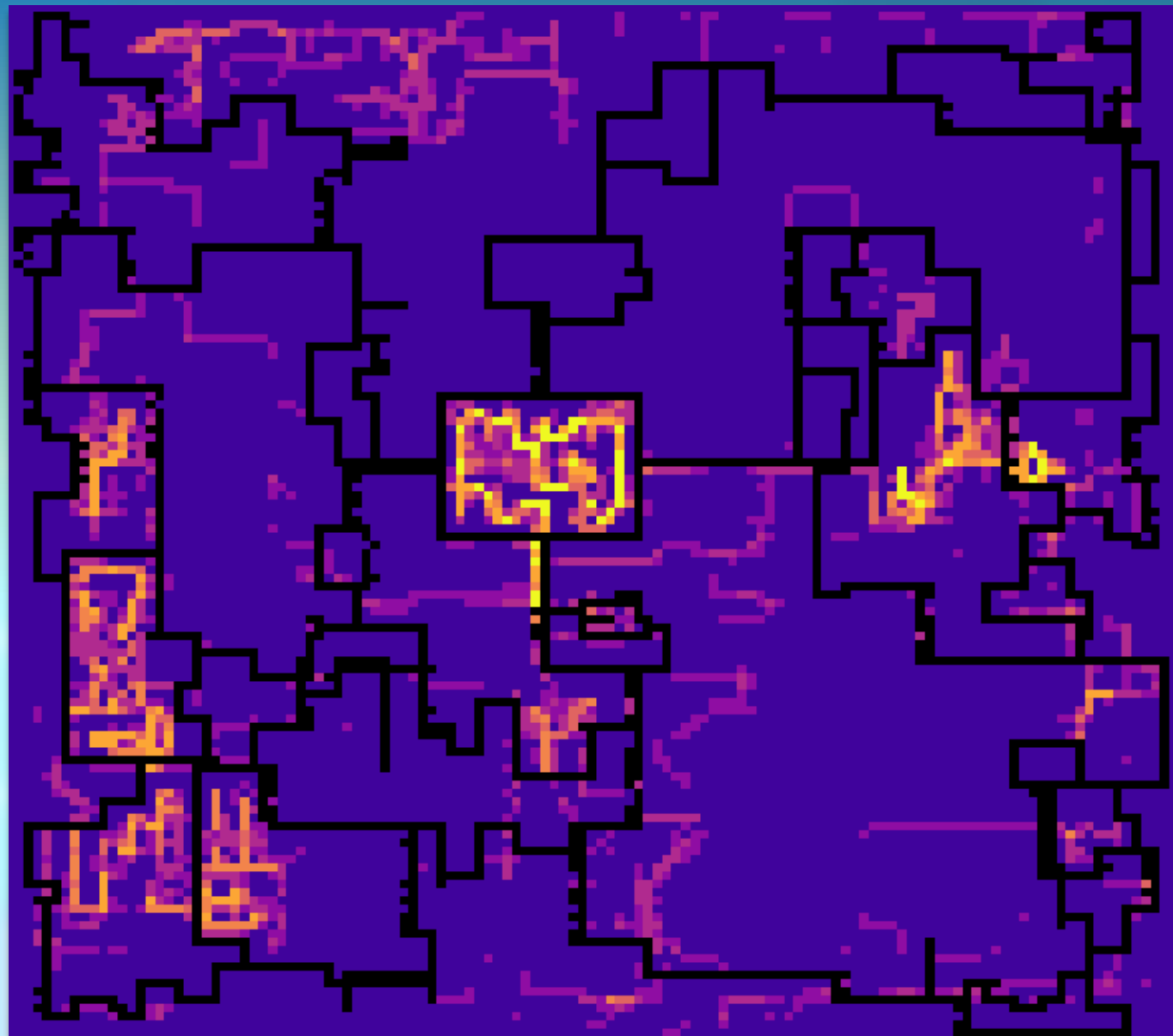
Data Collected

- . Telemetry
- . Questionnaire

MAIN STUDY



RESULTS



PATH USAGE

Navigation Help & Confusion

- . Footprints helped them find chests
- . Occasional confusion or clutter

FINDINGS

Sense of Community

- . world was co-created
- . environment felt 'populated' by prior visitors

FINDINGS

Exploration & Divergence

- triggered the opposite of what we expected
- footprints simultaneously served as shortcuts

FINDINGS

Creativity

- Rare but noteworthy



GAME REVIEWS REVIEWED:

A GAME DESIGNER'S PERSPECTIVE ON AI-GENERATED GAME REVIEW ANALYSES

Michael Lankes & Andreas Stöckl
COG 2023



UNIVERSITY
OF APPLIED SCIENCES
UPPER AUSTRIA

Contextualizing the Gaming Landscape

- *Gaming industry: exponential growth in the past decade*
- *Influx of player feedback in the form of reviews*
- *Challenge: analysis due to their sheer volume*

MOTIVATION



The Significance of Game Reviews

- *Direct channel of communication: players and developers*
- *Feedback: gameplay mechanics to narrative structures*
- *Iterative game design processes could be guided*

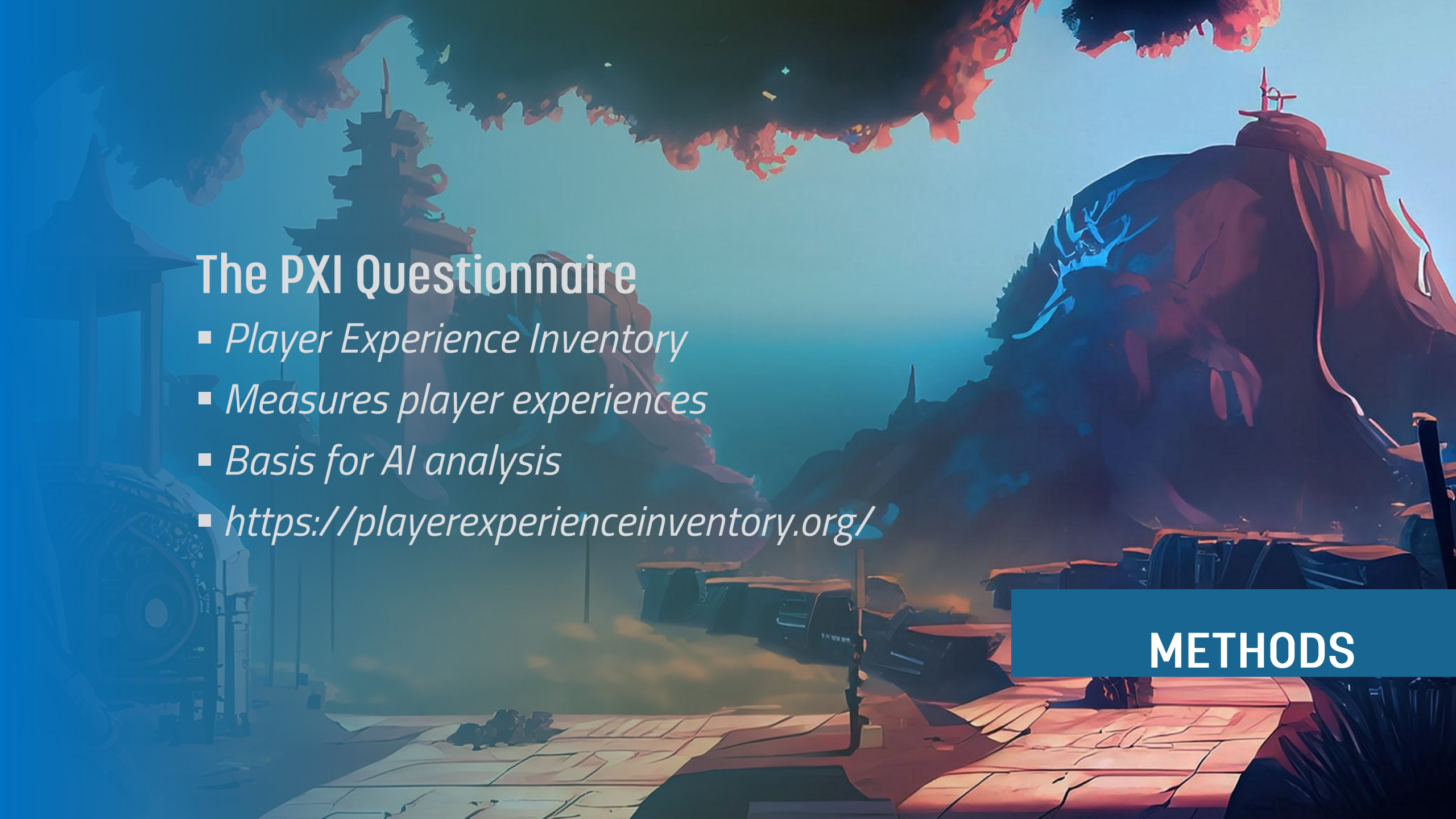
MOTIVATION



The GPT-3 Model & Its Capabilities

- *GPT-3's architecture allows it to understand context*
- *Process and generate human-like text*

CONTEXT



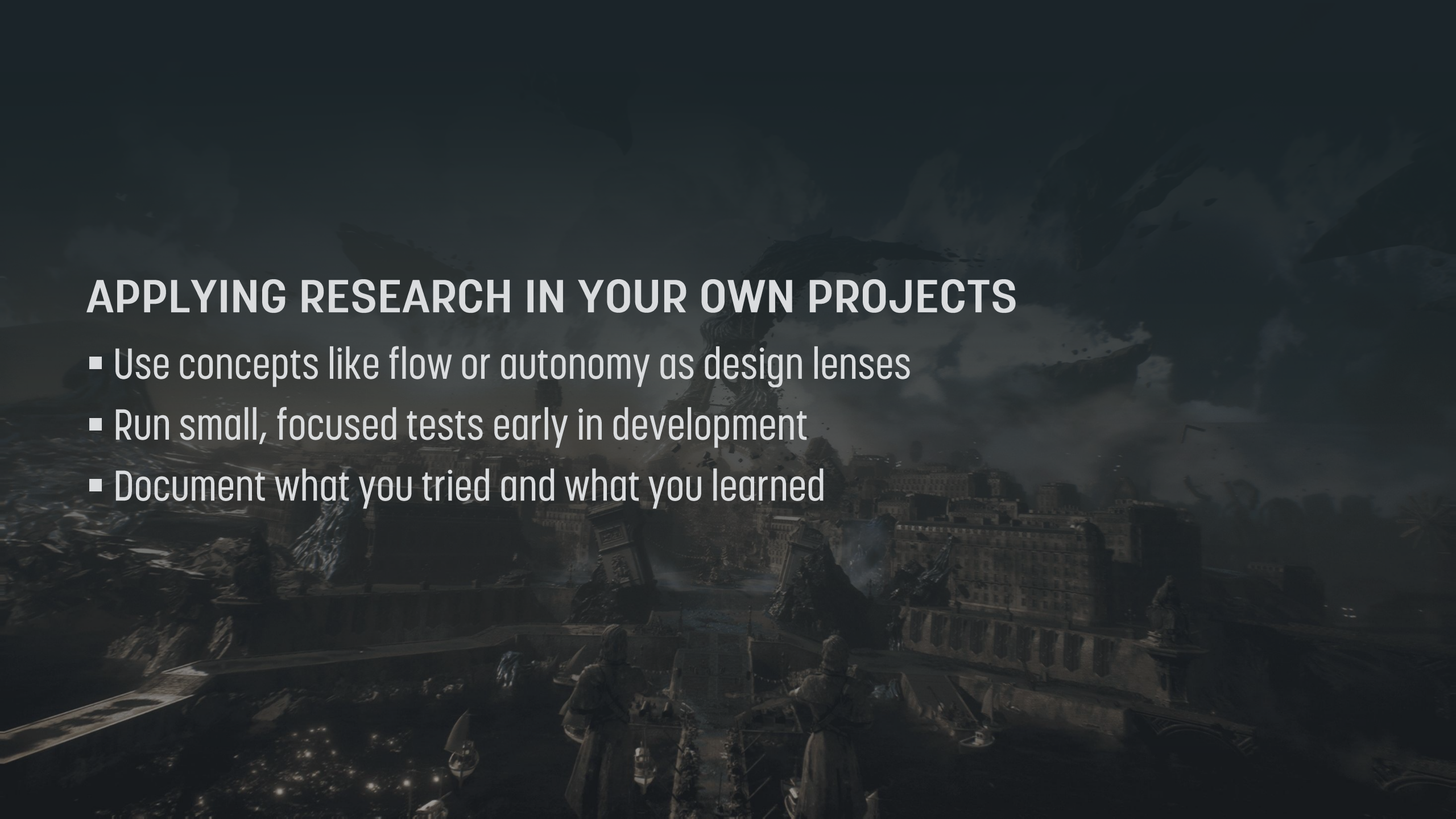
The PXI Questionnaire

- *Player Experience Inventory*
- *Measures player experiences*
- *Basis for AI analysis*
- *<https://playerexperienceinventory.org/>*

METHODS

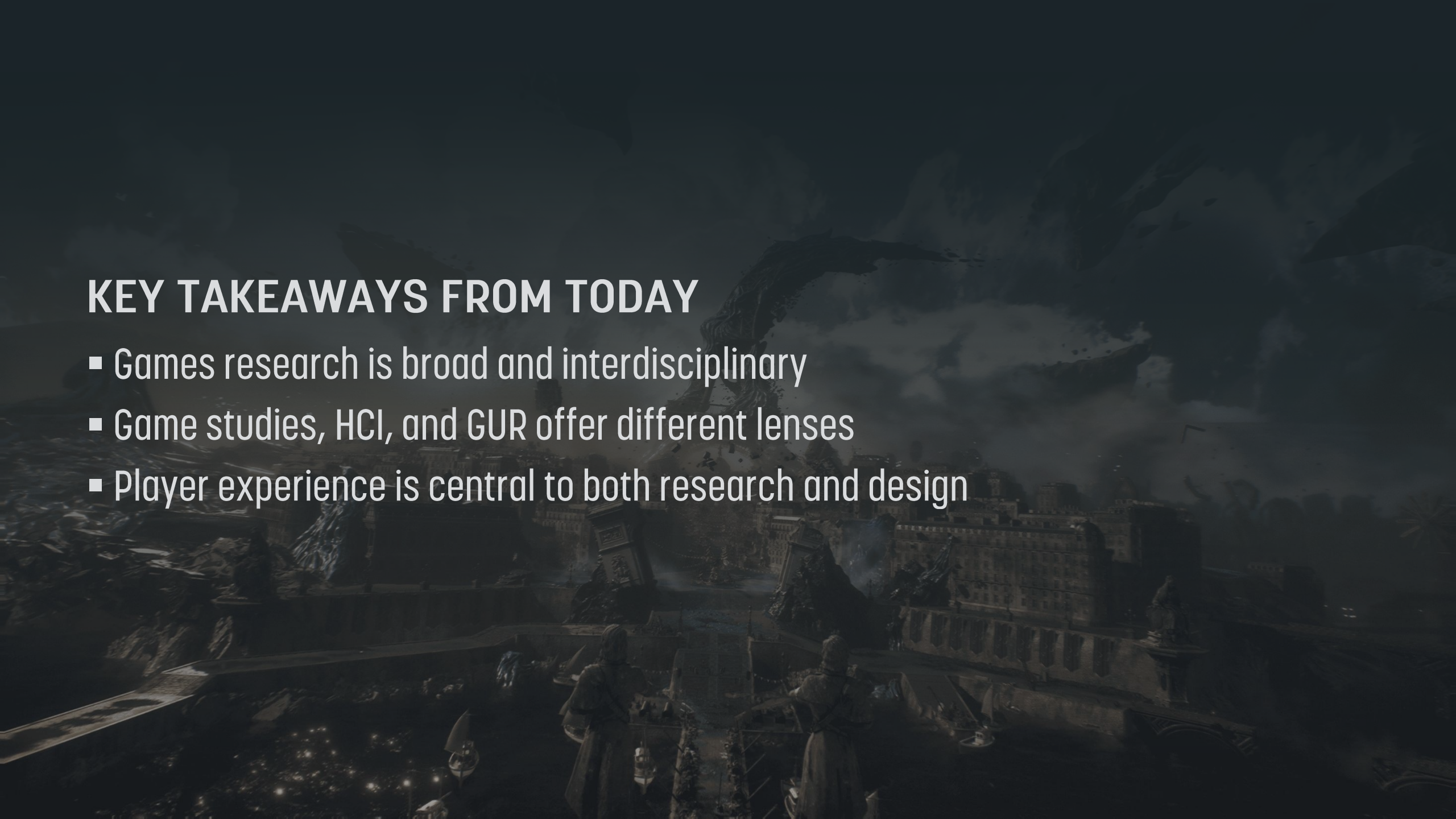
APPLYING RESEARCH IN YOUR OWN PROJECTS

- Use concepts like flow or autonomy as design lenses
- Run small, focused tests early in development
- Document what you tried and what you learned



KEY TAKEAWAYS FROM TODAY

- Games research is broad and interdisciplinary
- Game studies, HCI, and GUR offer different lenses
- Player experience is central to both research and design



HOW RESEARCH SUPPORTS BETTER GAMES

- Reveals hidden problems and opportunities
- Gives language to discuss experience and meaning
- Helps teams make more informed design choices



THANK YOU FOR YOUR ATTENTION!

